

Pharmacoeconomics

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Summary of PECO methods

Method	Description	Application	Cost unit	Output unit
COI	Estimate the cost of disease on defined population	Used to provide baseline to compare prevention/treatment options	\$\$	NA
CMA	Finds the least expensive cost alternative	Used when benefits are the same	\$\$	Assumed to be equivalent
CBA	Measures benefits in monetary unit and computes net gain	Can compare programmes with different objectives	\$\$	\$\$
CEA	Compares therapeutic effects measured in physical units	Can compare therapies/programmes that differ in clinical outcome and use same unit of benefit	\$\$	Natural unit

Summary of PEco methods

Method	Description	Application	Cost unit	Output unit
CUA	Measures therapeutic consequences in utility units rather than physical units	Used to compare medicines/programmes that extend life and have serious side effects or those provides reduction in morbidity	\$\$	QALYs/ DALYs
CCA	Measure multiple costs and outcomes without aggregating in to CE or CB ratio	Examine whether the use of a drug produces an outcome that decreases costs and offset price cost of a new therapy	\$\$	Reported separately

Outcome Measures

<i>Disease</i>	<i>Indicator</i>	<i>Clinical Outcome</i>	<i>Humanistic Outcome</i>	<i>Economic Outcome</i>
<i>Hypertension</i>				
<i>Hyperlipidemia</i>				
<i>Diabetes</i>				
<i>Asthma</i>				

Outcome Measures Example

<i>Disease</i>	<i>Indicator</i>	<i>Clinical Outcome</i>	<i>Humanistic Outcome</i>	<i>Economic Outcome</i>
<i>Hypertension</i>	BP	Renal failure Stroke MI Death	QOL	Cost/↓ mmHg BP Cost/stroke avoided Cost/life year saved
<i>Hyperlipidemia</i>	LDL levels	Angina MI Death	QOL	Cost/MI avoided Cost/point ↓ in LDL
<i>Diabetes</i>	A1C, BG levels	Retinopathy nephropathy Death	QOL	Cost/kidney transplant avoided
<i>Asthma</i>	FEV, peak flow	Exacerbation event Death	QOL	Cost/symptom free day

Steps in Pharmacoeconomic Evaluation

1. Establish the framework for the analysis
2. Determine the study design
3. Select the effectiveness measure and method for obtaining the data.
4. Identify relevant resources and measure usage, select unit prices, and estimate costs in each year.

5. Discount costs and effects in each year in the future and
sum over all years in time horizon
6. Compute an incremental decision index
7. Conduct sensitivity and/or statistical analyses to
determine the influence of uncertainty on conclusions
8. Interpret the cost-effectiveness results

1. Establishing the Framework

Includes

- Defining **the problem**
- Identifying the **perspective** and **alternative** interventions to be compared
- Identifying and **measure outcomes** of each alternative

Framework...

- Establishing the Framework take account of **decision context** and **Study Boundaries**

Decision Context

- Perspective , Target Population and Comparators

Study Boundaries

- Groups of Persons, Types of Outcomes and Time Horizon

Decision or Problem Context

- What is the issue/problem/question (economic dilemma) that bears addressing?
- Who are the likely stakeholders and what are their economic issues?

The context guides the choice of **study perspective, target population, and comparator.**

PERSPECTIVES

- The importance of pharmacoeconomic information to healthcare decision makers will depend upon the viewpoint from which the analysis is conducted.
- Perspective is **the point of view from which a pharmacoeconomic analysis is conducted**
 - **Providers** (physicians, pharmacists, nurses),
Administrators, Payers (insurance companies),
Employers, Patients, Society.

- Assessing costs and consequences—the value of a pharmaceutical product or **service—depends heavily on the perspective of the evaluation.**
- A PE evaluation can assess the value of a product or service **from single or multiple perspectives.**
- How much a drug costs depends on your point of view.

- **From a patient's perspective out-of-pocket expenses are the costs;**
- **To a hospital pharmacy department it may be wholesale drug purchase prices;**
- **To a national drug subsidy program it may be the costs of drugs (including containers) plus professional dispensing fees minus any patient contributions via prescription charges;**

PERSPECTIVES

- **To a national health program it may be the costs of the drug (+) the costs of appropriate monitoring of therapy minus the savings in a reduced number of health outcomes that would incur costs.**
- **From a societal perspective, there may be additional considerations— the costs to careers, time lost from work due to treatment, illness or death, and travel costs to attend clinics or general practitioners.**

PERSPECTIVES...

- Perspective asks from whose point of view is the study conducted - from that of the **health care payer, who is only interested in the direct costs of health care, or from society as a whole, where “indirect” costs (i.e. not directly on health care, such as lost of productivity etc) are also important.**
- Clarification of the perspective is critical because the results of a pharmacoeconomic evaluation depend heavily **on the perspective taken.**

PERSPECTIVES

- Once the perspective is clear, a full evaluation of the relevant costs and consequences can begin.
- Again, perspective is critical because **the value placed on a treatment alternative will be dependent heavily on the point of view taken.**

[What should be the perspective of a pharmacoeconomic analysis?]

- **A societal perspective is the most comprehensive, as it includes all costs and benefits irrespective of who pays and who benefits,** but often more limited perspectives are adopted.

The important point is that the perspective of the analysis needs to be clearly stated. Usually,

- The perspective that determines the range of costs (resources used) should needs to be considered.
- In general, the societal perspective is considered the most appropriate, **but a health care manager with a limited budget might be tempted to ignore the societal view and consider only the costs that fall on his own budget**, therefore might be considered as health service perspective.

Perspective

Common Perspectives

- Societal
- Third-party payer
- Institutional
- Patient

Importance:

- Perspective **affects the scope of costs** assessed

1. Patient Perspective

- Can you think of direct and indirect costs that a patient would pay for related to a asthma service?

Patient Perspective...

- Costs from the perspective of patients are essentially **what patients pay for a product or service, that is, the portion not covered by insurance.**
- Consequences, from a patient's perspective, are the **clinical effects, both positive and negative, of a program or treatment alternative.**

Patient Perspective...

- For example, various costs from a patient's perspective might include **insurance copayments and out-of-pocket drug costs, as well as indirect costs, such as lost wages.**
- This perspective should be considered when assessing the **impact of drug therapy on quality of life or if a patient will pay out-of-pocket expenses for a healthcare service.**

1. Patient Perspective



Asthma management
service

- Direct
 - Insurance premium
 - Transportation
 - Prescription
- Indirect
 - missed work

2. Provider Perspective

- **Costs from the provider's perspective are the actual expense of providing a product or service**
- **Providers can be hospitals, managed-care organizations (MCOs), or private-practice physicians.**
- **From this perspective, direct costs such as drugs, hospitalization, laboratory tests, supplies, and salaries of healthcare professionals can be identified, measured, and compared.**

Provider Perspective...

- However, **indirect costs can be of less importance to the provider.**
- When making formulary management or drug-use policy decisions,
 - the viewpoint of the healthcare organization should dominate.

Institutional perspective

Community pharmacy providing
asthma management services



- Direct
 - Equipment
 - Fees
 - Education
 - Marketing
- Indirect (overhead / accounting costs)
 - Maintenance
 - Utilities
 - Business licenses
 - Laundry, waste disposal e.t.c.

3. Payer Perspective

- Payers include **insurance companies, employers, or the government.**
- From this perspective, **costs represent the charges for healthcare products and services allowed, or reimbursed, by the payer.**
- The primary cost for a payer is **of a direct nature.**
 - However, **indirect costs, such as lost workdays and decreased productivity,** also can contribute to the total cost of healthcare to the payer.
- When insurance companies and employers are contracting or selecting healthcare benefits for their employees, then the **payer's perspective should be employed.**

Payer Perspective...

- Direct

Hospitalization

Laboratory costs

Medication

Medical devices

Physicians /pharmacist fees

- Indirect ?

4. Societal Perspective

- The perspective of society is the **broadest** of all perspectives because it is the only one that considers the benefit to society as a whole.
- Theoretically, **all direct and indirect costs are included** in an economic evaluation performed from a societal perspective.

Societal Perspective...

- Costs from this perspective include **patient morbidity and mortality and the overall costs of giving and receiving medical care.**
- An evaluation from this perspective also would include **all the important consequences an individual could experience.**
- In countries with nationalized medicine, like Ethiopia, **society is the predominant perspective.**

Societal Perspective...

- Asthma cost of illness



Direct

- Health care costs
- Taxes,
- medication research and development ,
- health care work force

Indirect

- Lost productivity
- Lost wages

Methods of economic evaluation

- All economic evaluation has common structure which involves explicit measurement of inputs(costs) and outcomes.
- The common issues in pharmacoeconomic evaluation are:
 - **Perspectives**
 - **Costs**
 - **Outcomes (consequences, benefits)**

Perspectives

Patient

- Clinical Care
- Quality of life
- Out-of-pocket Cost
- Satisfaction with treatment process

3rd-Party Payer

- Clinical Cure
- Cost
- Customer perception of value

Hospital / Physician

- Clinical Cure
- Profit from treatment

Employer / Society

- Clinical Cure
- Cost
- Productivity



CHAPTER THREE

Decision Analysis and Pharmacoeconomic Evaluations

3.1. Basic concept of Probability

- Probability is a numerical measure of the likelihood that an event will occur.
- Probability of an event $P(E)$: “Chance” that an event will occur
- Economic models are often probabilistic
- Quantifying uncertainty for purposes of:
 - Prediction
 - Assessing preferred course of action

Why study probability?

- Inferential statistics—inferences about population from:
 - Sample
 - Experiment (or random experiment)

...Basic concept of Probability...

Types of Probability

A. Objective probability:

i. Relative Frequency Approach: Relative frequency of an event occurring in an infinitely large number of trials

Time Period	Number of Male Live Births	Total Number of Live Births	Relative Frequency of Live Male Birth
1965	1927.054	3760.358	0.51247
1965-1969	9219.202	17989.360	0.51248
1965-1974	17857.860	34832.050	0.51268

ii. Equally-likely Approach: If an experiment must result in n equally likely outcomes, then each possible outcome must have probability $1/n$ of occurring.

Examples: 1. Roll a fair die 2. Tossing a coin

B. Subjective Probability: A number between 0 and 1 that reflects a person's degree of belief that an event will occur; Example: Predictions for rain

...Basic concept of Probability...

Definition

The frequentist school of statistics

- The probability of an event **as the number of times the event occurs divided by the number of trials in which it could have occurred** ,n, as n approaches infinity.
- $P(A) = \frac{\text{[The number of times the outcome occurs]}}{\text{[Total number of trials]}}$

- For example, the probability that a coin will come up heads is 0.5 because, assuming the coin is fair, as the number of trials (flips of the coin) gets larger and larger, the observed proportion will be, on average, closer and closer to 0.5.
- Problem:
 - Repetition of events may not be infinite. Some events can happen only once

....Basic concept of Probability...

The Bayesian school

- The probability of any event occurring **as the personal degree of belief that the event will occur.**
- Therefore, if I personally believe that there is a 70% chance that team A will win tonight's game, then that is my probability for this event.
- Problem: statements reflect **personal subjective beliefs**

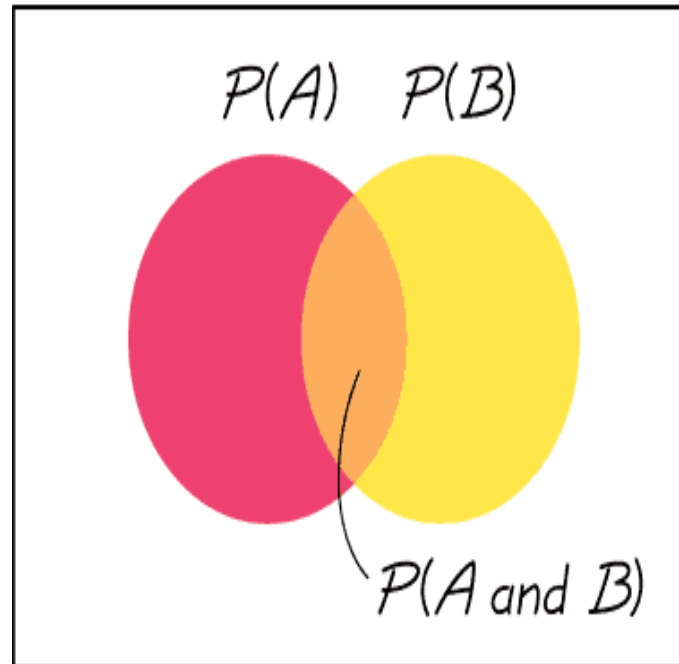
The Bayesian school...

- Having a range of probabilities depending on the personal beliefs of a community of clinicians is a useful reflection of reality.
- As more data accumulate, Bayesian and frequentists probabilities tend to agree, each essentially converging to the mean of the data. When this occurs, similar inferences will be reached from either viewpoint.

....Basic concept of Probability...

Concept of Union, Intersection and Complement

- Pink + Gray + yellow = $A \cup B$
- Gray = $A \cap B$
- Yellow = A'
- Pink = B'



....Basic concept of Probability...

Unions of Two Events

- “If A and B are events, then the union of A and B, denoted by $A \cup B$, represents the event composed of all basic outcomes in A or B.”
The notation for the union is $P(A \cup B)$.

Intersections of Two Events

- “If A and B are events, then the intersection of A and B, denoted by $A \cap B$, represents the event composed of all basic outcomes in A and B.” The notation for the intersection is $P(A \cap B)$.

Law of Complements:

$$P(A') = 1 - P(A)$$

- The complement (A') of an event A is the set of all sample points that are not in event A.

$$P(A) + P(A') = 1 \text{ or } P(A') = 1 - P(A).$$

....Basic concept of Probability...

Rules of probability

Four basic rules of probability exist

Rule 1

- By convention, all probabilities are numbers **between 0 and 1**.
- The value of a probability can only be $0 \leq p \leq 1$
- A probability of **0 indicates an impossible** event, and a probability of **1 indicates an event certain to happen**.
- Most events of interest have probabilities that fall between these extremes.

Rule 2. Mutually exclusive events and the additive rule of probability

- Events are termed **disjoint (exclusive)** if they have no outcomes in common.
- Two or more events are said to be mutually exclusive if at most one of them can occur when the experiment is performed, that is, **if no two of them have outcomes in common**
- If two events are mutually exclusive (disjoint), the probability that one or the other will occur equals the sum of the probabilities:

$$-p(A \text{ or } B) = p(A) + p(B)$$

Basic concept of Probability...

- Example- Role a six sided Die. The possible outcomes (Sampling space) are six (1,2,3,4,5,6). Each event has equal probability of occurrence (i.e. $1/6$).
- Probability of rolling an even number would be:
 - $p(\text{even}) = p(2) + p(4) + p(6)$
 - $= (1/6) + (1/6) + (1/6) = 1/2$
- What if A and B are two events, not necessarily disjoint, then
 - $p(A \text{ or } B) = p(A) + p(B) - p(A \text{ and } B)$

- The probability that at least one event occurs is the probability of one event plus the probability of the other. **But to avoid double counting, the probability of the intersection of the two events is subtracted.**
- Example: At Pharma college, all first-year students must take Anatomy and Physiology. Suppose 15% fail Anatomy, 12% fail Physiology, and 5% fail both. Suppose a first-year student is selected at random. What is the probability that student selected failed at least one of the courses?

....Basic concept of Probability...

Rule 3.

- The sum of the probabilities that an event will occur and that it will not occur is equal to 1.

Rule 4. Independent events and the multiplicative rule of probability

- For two given events, if the occurrence or nonoccurrence of one doesn't affect in any way the occurrence or nonoccurrence of the other, the events are called independent events (i.e., knowing the outcome of one provides no information concerning the likelihood that the other will occur)

Independent events...

- Then the probability that both events will occur is given by the product of their individual probabilities
 $p(A \text{ and } B) = p(A)p(B)$

Example

- If the probability of diagnosing a malignant tumor is 0.1, and the probability that it will rain today is 0.3 (an independent event) then the probability of a malignant tumor and rain today is
 $0.1 \times 0.3 = 0.03$, or 3%

Conditional probability

- The probability that **event B occurs given that event A has occurred** is called a **conditional probability**.

Example: What is the probability that a neonate died? Clearly, this depends on a number of factors, including maternal wellbeing, gestational age, birth weight, the presence or absence of neonatal illness etc.

Conditional probability...

- The probability of neonatal death is conditional on other factors and is not a single constant number by itself.
- Such probabilities are known as **conditional probabilities**
- Many probability calculations used in health research involve conditional probabilities

Conditional probability...

- If unconditional probabilities can be denoted by $\Pr(\text{ neonatal death})$, then conditional probabilities can be denoted by; $\Pr(\text{ND} | \text{LBW}(<2500 \text{ gm}))$, read as “the probability of ND given or conditional on a LBW baby,” and similarly,

Conditional probability ...

- $\Pr(\text{ND} / \text{normal birth weight})$, read as “the probability of ND given a NBW baby”
- These probabilities are **highly relevant in practice** and research. Are also **useful especially in economic analysis** because **probabilities of an event differ, depending on other events occurring**

Conditional probability...

General Notations

- If we denote one event by A, and a second event by B, then
- $\Pr(A | B)$, is given by dividing the unconditional probability that these two events occur together by the unconditional probability that B occurs

$$\Pr(A|B) = P(A \cap B) / P(B)$$

- This formula is conveniently rewritten as the following which follows the **Multiplicative Rule**.

$$P(A \cap B) = P(A|B) \times P(B)$$

- **Example :** Assume that the overall probability of a patient in the clinic having endometrial cancer is 10%. Furthermore, suppose that the sensitivity of endovaginal sonography for diagnosing endometrial cancer is 90%. If we let A represent the event that the patient has a positive test result and B represent the probability of endometrial cancer in this patient population
—(i.e. $\Pr(B) = 0.1$ and $\Pr(A | B) = 0.9$.)

- Therefore, the probability that a patient in this clinic has both endometrial cancer and positive results is

$$-0.1 \times 0.9 = 0.09 \text{ or } 9\%$$

Learning Objectives

- Interpret and critique the structure of a decision tree that depicts a pharmacotherapy choice situation.
- Explain how expected costs and outcomes are calculated from a decision tree

Modeling

- A simplified version of reality that is used to describe the essential elements of a real situation

Decision Models

- Used to **piece together** the aggregate implications of diverse pieces of evidence from multiple data sources that bear upon a choice from alternative courses of action

Decision Analytic Modeling

- A technique used to **evaluate competing decisions**
- Can focus on cost, outcomes or both
- Uses a “decision tree” to help determine the best selection

- **Validity** of the model rests on whether its assumptions are reasonable in light of the needs and **purposes of the decision maker**, and in terms of the model's implications
- Synthesizes data from the **literature** to **model the disease, treatment and outcome process.**

3.2 Decision analysis

Definition and theoretical bases

- **A systematic approach to decision making under conditions of uncertainty**
- **A systematic Quantitative approach that assess the relative value of complex decision options with uncertain outcomes.**
- Provides a formal, transparent, and orderly analytic approach to assist the decision maker in identifying the preferred course of action from competing alternatives

- It helps in answering questions like
 - What is the effectiveness of a new treatment?
 - What is the chance of adverse event ?
 - What are the costs associated to treatment failure?
 - How cost-effective are treatment alternatives ?
 - What statistical distribution surrounds cost probabilities or outcomes?
 - What impact might non-adherence have on cost effectiveness?
 - Based on my threshold to pay for health care, is this cost effective option?

Decision analysis...

- Historically Decision Analysis (DA)was developed as a method to help clinicians make decisions on how to manage individual patients-as a quantitative clinical epidemiology tool for physicians who wished to quantify: – **Expected risks, Benefits, Utilities and Sometimes cost** associated with alternative treatment options.
- At its essence, it is a ‘simulated, mathematical, disease state model’

Decision analysis...

- May utilize *expert opinion, literature-based results, data from prospective or retrospective analyses*
- Provides the conceptual foundation for **cost-effectiveness, cost-utility, and cost-benefit** analyses
- This general set of methods was later adopted for **structuring and analyzing collective decisions in health care such as program evaluation and economic appraisal.**

Decision analysis...

- **When to use DA?:**
 - There are **real** alternatives,
 - There is **uncertainty**
 - When **consequences are important or have series consequences**

- **Information from DA can be used:**
 - **To decide how to manage an individual patient**
 - **To formulate policy recommendation about group of similar patients.**
- **DA yields an estimate of the net value of the different decision options in relation to each other**

Key Attributes

- **All relevant aspects of the decision and relevant strategies are explicitly articulated:**
 - Base Case
 - Perspective
 - Time Horizon
 - Treatment strategies
 - Probabilities
 - Costs
 - Outcomes
 - Uncertainty
- **Alternative course of action**
 - Novel treatments
 - Standard of care
 - Watchful waiting
- **Chance events**
 - Probability of treatment failure
 - Probability of adverse reaction
- **Consequences**
 - Economic, Clinical or Humanistic outcomes

...Decision analysis...cont

Typical Steps in Conducting a Decision Analysis

- 1) Establish the **research question**
- 2) Define the **perspective** (i.e., ‘Costs to whom’)
- 3) Define the **base case(s) of analysis** (e.g., patient and disease characteristics)
- 4) Specify **treatment modalities** and **choose appropriate comparator(s)**, e.g., least costly, standard of care, most-commonly used

...Decision analysis...cont

- 5) Model the disease state, define the time horizon, and choose final outcomes
- 6) Populate the **model with probabilities, costs, and outcome data**
- 7) Verify the model, calculate and report results
- 8) Conduct additional sensitivity analyses

Steps in Conducting a decision Analysis

- The **basic question should be framed and the perspective chosen,**
- the structure of the problem should be developed,
- the **probabilities and values for the outcomes should be estimated,**
- the tree should be analyzed to obtain the expected value of the outcomes, and
- **sensitivity analysis** should be conducted to evaluate the effect of assumptions on the results.

...Decision analysis...cont

Models used for Economic Evaluation (EE)

- What is a Model?
 - In the context of EE of medical intervention, a model is any mathematical structure that **represents the health and economic outcome of patients or population** under variety of scenarios.
 - Fundamental **analytic tools used in DA to display the temporal and logical sequence of decision problem**

Modeling versus Trial

- Trial based EE **are typically based not only on the data collected in CT.**
- Construction of model based entirely on **data from the literature and existing database** can often be the only or the best approach available
- Clinical trials aren't always **feasible b/c of ethical consideration or sample size requirements**

Decision analysis...

Types of Modeling Methods in DA

- Many methodologies and modeling types can be **used to create and evaluate decision problems**
- the method most appropriate to the particular problem should be used.

Types of modeling methods frequently used in health technology assessments are:

- Decision trees
- Markov models
- Simulation models
- Deterministic (Mechanistic) models

The choice is dependent upon:

- **The complexity of the problem,**
- **The need to model outcomes over extended periods of time, and**
- **Whether resource constraints and interactions of various elements in the model are required**

3.2.1. Decision Trees

- Decision analysis is a **quantitative, probabilistic** way of modeling problems **under situations of uncertainty**
- The decision tree is **Probably the most common structure** for decision models in EE.

Decision Trees...

- DA for EE proceeds by careful structuring of the problem using decision tree –a graphic scheme where we **begin with the decision and trace out all probable pathways and consequences**
- It`s a diagrammatic representation of the **possible outcomes and events used in DA**

Decision Trees...

- **It represents individual's possible progress, following some sort of intervention, by a series of pathways**
- **The sequential steps proceed with each step depending on the decision or probability outcome from the preceding step**

Decision Tree

- Depict potential health consequences as fixed pathways emanating from each alternative
 - Model is **linear**, i.e., forward flowing
 - Each *pathway terminus* has associated **cost & health outcomes**

- Shows the structure of a clinical situation, that is, the logical and temporal sequence of decisions and probabilistic events in the clinical situation.
- It also allows different outcomes to be weighted according to the value they have for the decision maker.

Decision Tree Elements

Structural Elements (structuring the problem)

- clinical starting point - target population or individual
- decision choices (decision nodes): treatments, tests, etc
(follow a square node)
- events (preceded by a chance node)

Data Elements

- probabilities of events
- values of outcomes – utilities (quality of life) or costs

Structuring the Problem

- Decision model is chosen to represent the **clinical problem**
- Model needs to be **simple** enough to be understood, but complex enough to capture the essentials of the problem
- Many **simplifying assumptions** are needed for modeling

Structuring the Problem: Decision Nodes

- **Decision node: point in time when a choice is made among competing strategies**
- Decision nodes are the **choices** being Considered for a problem
- These choices are under the control of the decision maker
 - To treat medically or surgically
 - To do something now or later
 - To do nothing at all

Structuring the Problem: *Chance Nodes*

- Chance nodes are the **temporal sequence of possible events** which follow a decision node
- Chance events are **out of the control of the decision maker**

(they're based on probability or outcome data)
- Chance events at a particular node need to add up to 100%

Chance Nodes

- **Objective data** -- based on hard data **from the literature** (randomized controlled trials)
- **Subjective data** -- softer data based on **expert opinion**
- Bayes' Rule applied to test results – predictive value of positive/negative tests

Structuring the Problem: Terminal Nodes

Terminal nodes:

- **terminal branches** of tree that represent **outcomes of a strategy**
- are the final outcome state associated with each possible pathway
- Some measure of value or worth needs to be applied to the terminal nodes
- Whose values: patient's, society, hospital, etc.
- Outcome measure needs to be consistent

Elements of a Decision Tree

- Event branches
- Nodes
 - Decision 
 - Chance (event) 
 - Terminal 
- Probabilities
- Rollback values

Rules of construction

- Tree flows from left to right
- Each chance node is followed by a set of mutually exclusive outcomes or events, depicted as branches on the tree
- Categories of outcomes/events must be chosen so that they are exhaustive (i.e., represent all possible outcomes)
- If above 2 conditions are met, probabilities after each chance node must sum to 1

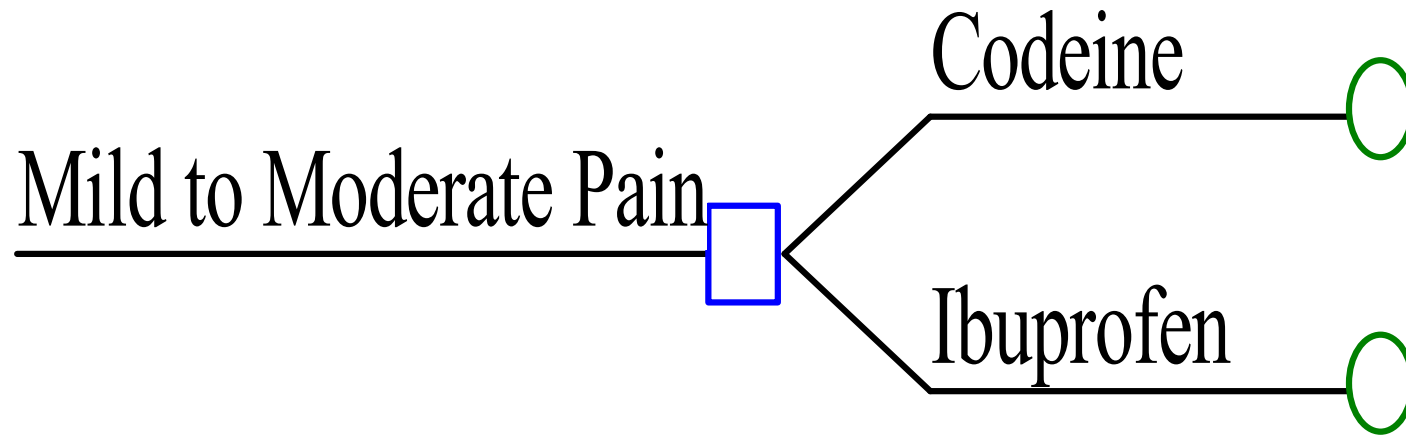
Constructing a tree: an example

Clinical Situation: acute, mild to moderate dental pain

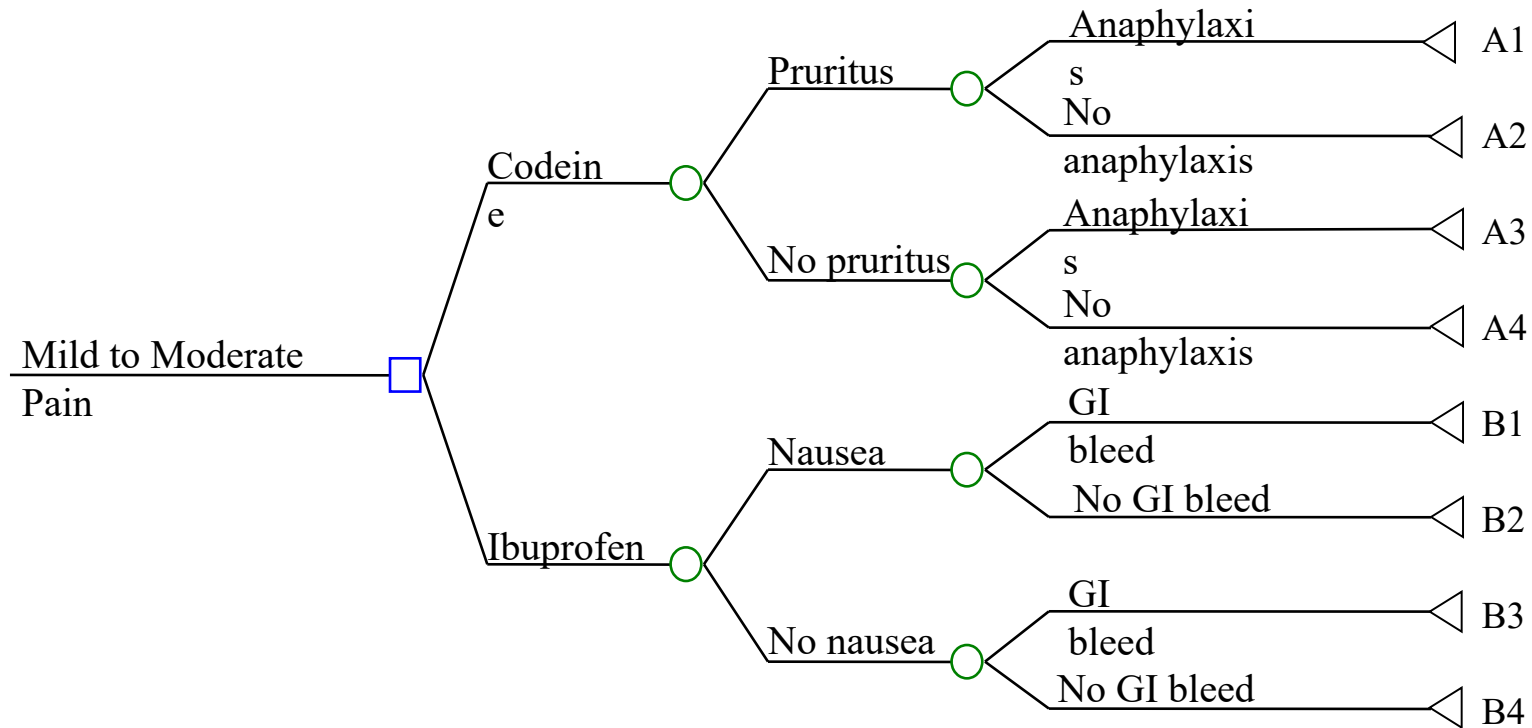
Choice: Codeine or Ibuprofen

Attributes: equally effective but differ in cost and side effects

Depicting the decision alternatives



Adding the Consequences



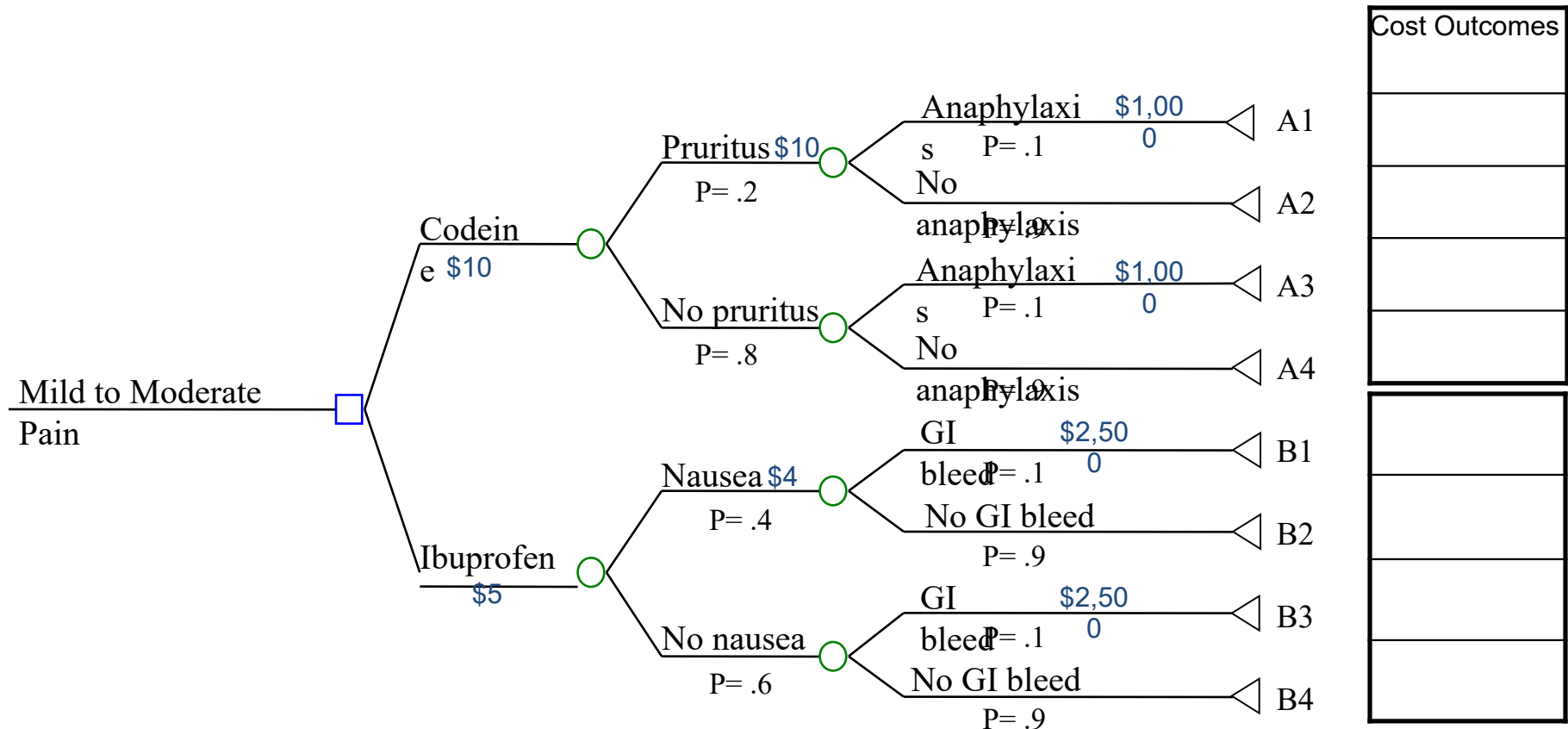
Populating” the decision tree

- Branches after each chance node assigned a probability, obtained by **converting event rates** from published studies
- Select probabilities from best quality data source:
 - Meta-analysis
 - Large RCT
 - Nonrandomized controls
 - Small RCT
 - Expert opinion

Importance of conditional probabilities

- The probabilities that are entered after each chance node, beyond the first one, are conditional probabilities:
 - The probability of an event B, given that an event A is known to have occurred

“Populating” the decision tree



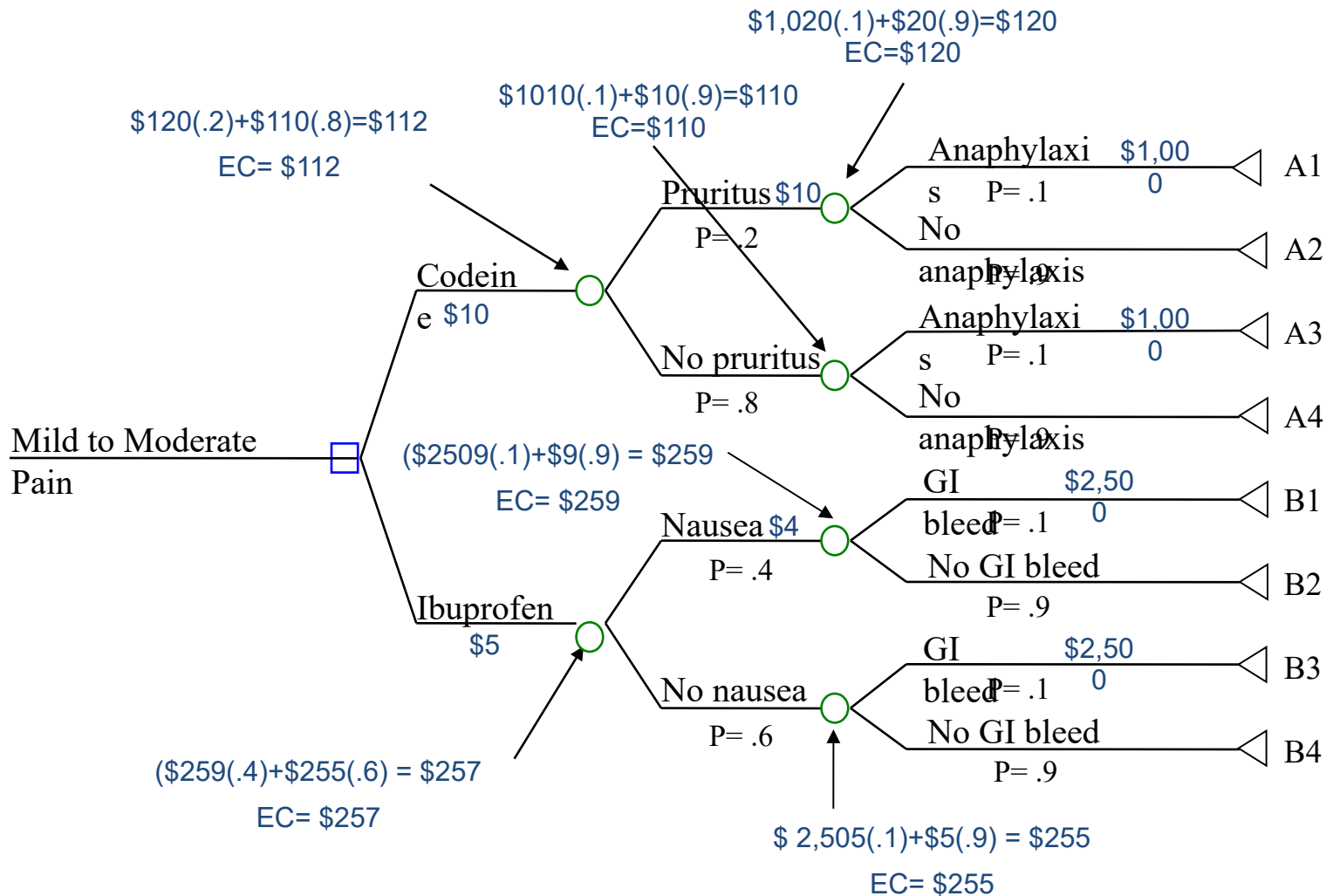
Calculating expected outcomes

- “Expected” means average, i.e., a probability-weighted outcome
- Calculate expected value at each node, proceeding from right to left. This is called “folding back” the tree.

$$\text{Expected Value}_{\text{node}_x} = \sum p_i V_i$$

- Do this separately for costs and quantified health outcomes, e.g., years of life, QALYs
- Do this for each choice alternative

Computing the decision tree



Cost Outcomes
\$1,020
\$20
\$1,010
\$10

\$2,509
\$9
\$2,505
\$5

Expected costs for codeine and Ibuprofen

- Codeine \$112; Ibuprofen \$257
 - ✓ Compare this to the cost of the drug alone.
 - ✓ What event and type of resource utilization was the main cost driver?
- Which treatment alternative is preferred?

Decision Tree-Example

- You have been asked to compare two treatments options, Drug A is an agent that has been in the market for over 5 years and drug B is a new agent recently approved for prescription. Using the data below employ decision analysis techniques to compare the cost effectiveness of the two options.

Parameters	drug A	Drug B
Cure rate	70%	90%
Significant adverse event rate	20%	10%
Cost per course of treatment	\$ 100	\$ 150
Cost to treat adverse event	\$ 25	\$ 25
Cost of treatment failure	\$ 200	\$ 200

Example

- Steps in this ‘deterministic’ example
- To obtain the
 - Structure of the decision tree
 - Path probabilities
 - Path costs
 - Average costs of Drug A and Drug B
 - Average effectiveness of Drug A and Drug B
 - Average cost-effectiveness of Drug A and Drug B
 - Incremental cost-effectiveness of the Drug B versus Drug A

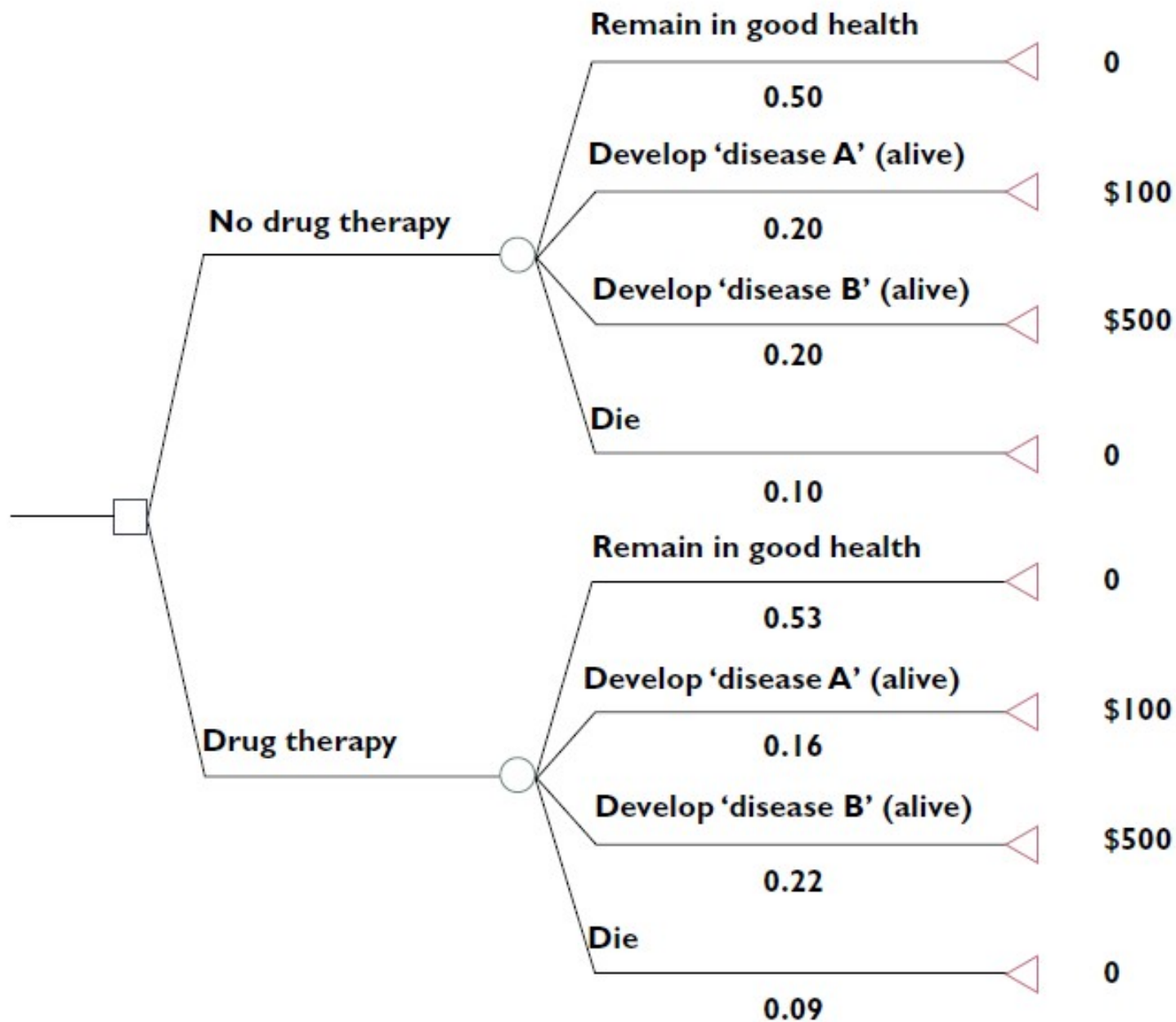
1. The expected value of 'No drug therapy' is? (0.78 utility) =

$$- 0.50 * 1.0 + 0.20 * 0.80 + 0.20 * 0.60 + 0.10 * 0)$$

2. the expected value of 'Drug therapy' is (0.79 utility)

$$- (0.53 * 1.0 + 0.16 * 0.80 + 0.22 * 0.60 + 0.09 * 0).$$

- Therefore on average, drug therapy would provide a (slightly) more favorable health return compared with no drug therapy.
- The consequences of an option can also be quantified in terms of costs. In Figure 2



3. Calculate expected cost for drug and non drug therapy?

3.1. Drug therapy = $E(\text{cost} \times \text{transition probability})$.

$$- \$126 = (0.53 \times \$0 + 0.16 \times \$100 + 0.22 \times \$500 + 0.09 \times \$0)$$

3.2. 'No drug therapy'

$$- \$120 = (0.50 \times \$0 + 0.20 \times \$100 + 0.20 \times \$500 + 0.10 \times \$0)$$

- On average, drug therapy would lead to greater downstream cost, even though it is associated with improved health outcomes.

3.3. If the cost of the drug itself is assumed to be \$300, then the *net cost of drug* therapy would be?

$$- \$300 + (\$126 - \$120) = \$306.$$

- On average, if drug therapy was to be delivered, it would **increase utility by 0.01** (from 0.78 to 0.79), but cost \$306 dollars more.
- The net cost per unit of benefit gained from intervention is the ***incremental cost-effectiveness ratio (ICER)***, which is the main outcome of interest in cost-effectiveness, including pharmacoeconomic, analyses.
- In the example, the ICER would be **\$306/0.01** utility gained = \$30 600 per utility gained.

- **Utilities** are multiplied by the **years** to which they apply to derive *quality-adjusted life years (QALYs)*
- In the example above, if it is assumed that the utilities stayed constant for 1 year (that is, the health benefit procured by drug therapy lasted for a whole year), then drug therapy would lead to
 - **0.01 QALY** gained (0.01 utility multiplied by 1 year), and the ICER would be **\$30 600** per QALY gained over 1 year.

Outline

- Summary of modeling approach
- What is Unique about a modeling approach to Pharmacoeconomic evaluation
- Pros and cons of the Modeling design
- Markov Models in CEA
- Advantages and disadvantages of Markov modelling
- How to measure quality of life?

Putting it all together in a cost-effectiveness analysis

- For each choice alternative have calculated **expected costs and expected effectiveness**
- Proceed to calculate incremental cost-effectiveness ratios, conduct sensitivity analyses, etc.
- Computer software
 - Excel
 - Decision analysis software

Summary of modeling approach

1. Structure the tree/build the model
2. Populate the tree/model for each alternative
3. Compute expected costs and health effects for
each alternative

What's different about a modeling approach to Pharmacoeconomic evaluation

- Effectiveness is estimated using data from **multiple sources (populations), collected using different methods (study designs)**

- Validity of expected effectiveness estimate depends on the validity of both **the model structure** and **the data used**
- There are more sources of uncertainty; therefore more extensive sensitivity analyses are typically required

Pros and cons of the Modeling design

PROS	CONS
Timely	Must make simplifying assumptions about model structure
Inexpensive	Dependent on availability of data in the literature or in databases
Can use epidemiological data to extrapolate beyond time horizon of clinical trial	Model splices together data from different populations and study designs

- Basic decision analysis considers **only one sequence of events within one time-frame**. This is a limitation if the conditions being simulated have different sequential stages, and/or data inputs (such as transition probabilities or costs) evolve with time.
- To overcome this inherent limitation of basic decision analysis, ***Markov modelling*** is often employed.

Markov Modelling

Markov Models in CEA

- A model is a hypothetical description of a system

Why not experiment with the actual system?

- If it is possible to experiment with the actual system, do it!
- However, especially in healthcare, it is frequently too expensive, too disruptive, too slow, and/or unethical to experiment with the actual system

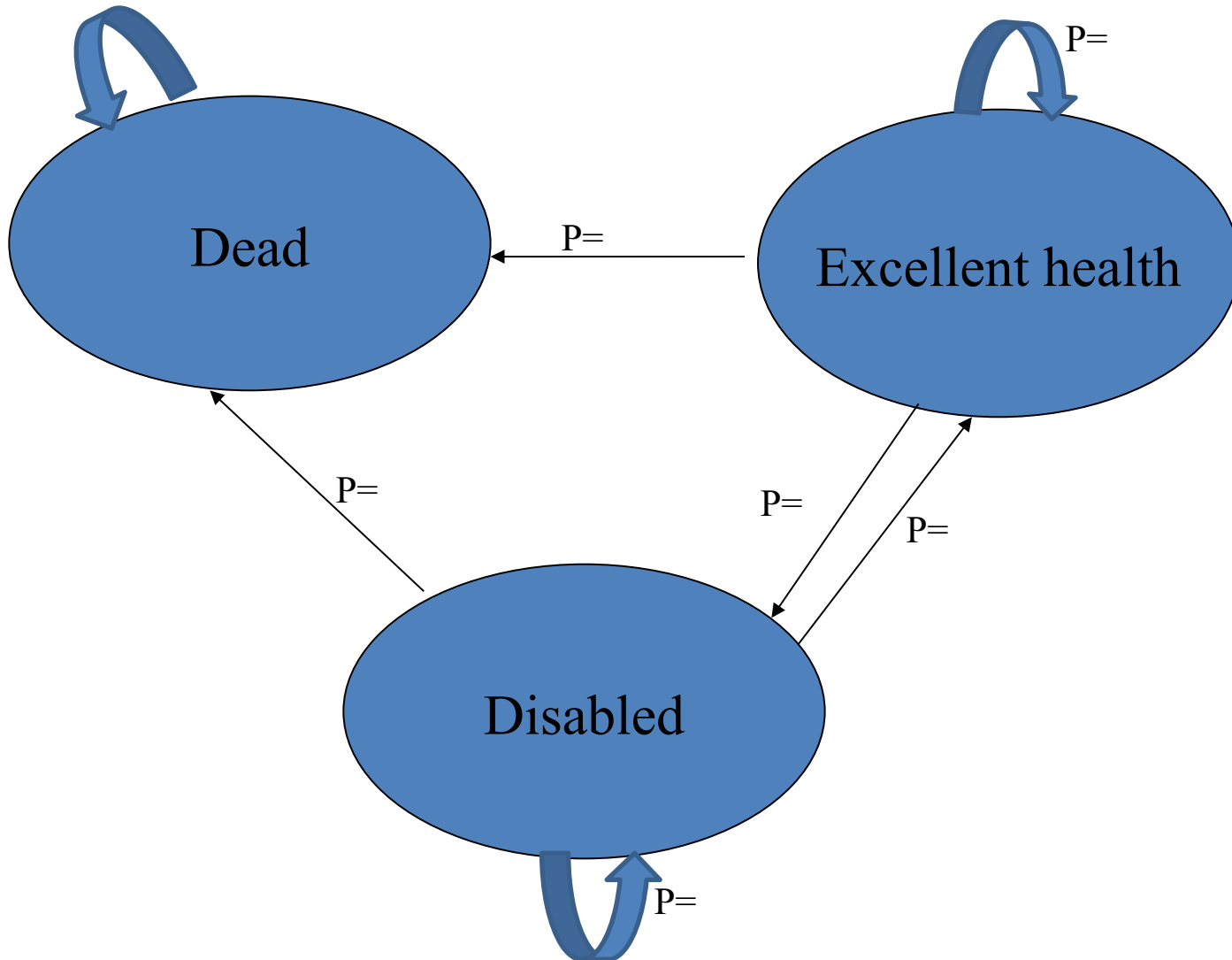
Markov Models in CEA

- depict potential health consequences as a **collective of health states** that patients could occupy at any defined point in time
- Useful for diseases in which events may occur **repeatedly over time**

- **Each *state*** has an associated cost and quality of life (utility)
- Recurring events accommodated by permitting **cycling back** to a previous state (recursive)
- Time horizon is divided into discrete increments called **cycles**

- Common representation of a simple Markov process called a **state-transition diagram**
- Each state represented **by a circle**
- **Arrows** connecting different states indicate **allowed transitions**
- States with arrows to itself indicate that patient may **remain in that state** in consecutive cycles
- **no transition** from “DISABLED” to “WELL”, nor “DEAD” to any other state

State Transition Model



Markov Models ...

- Markov processes are categorized by whether or not state-transition probabilities are constant over time
- For example, transition probability from WELL to DEAD may consist of two components:
 - Probability of dying **unrelated to disease** in question – changes over time as patient gets older
 - **Probability from disease** (e.g., fatal hemorrhage or embolus during cycle) may or may not be constant over time

- **A sensitivity analysis** allows one to determine how the results of an analysis would change when these best guesses or assumptions are varied over a relevant range of values.

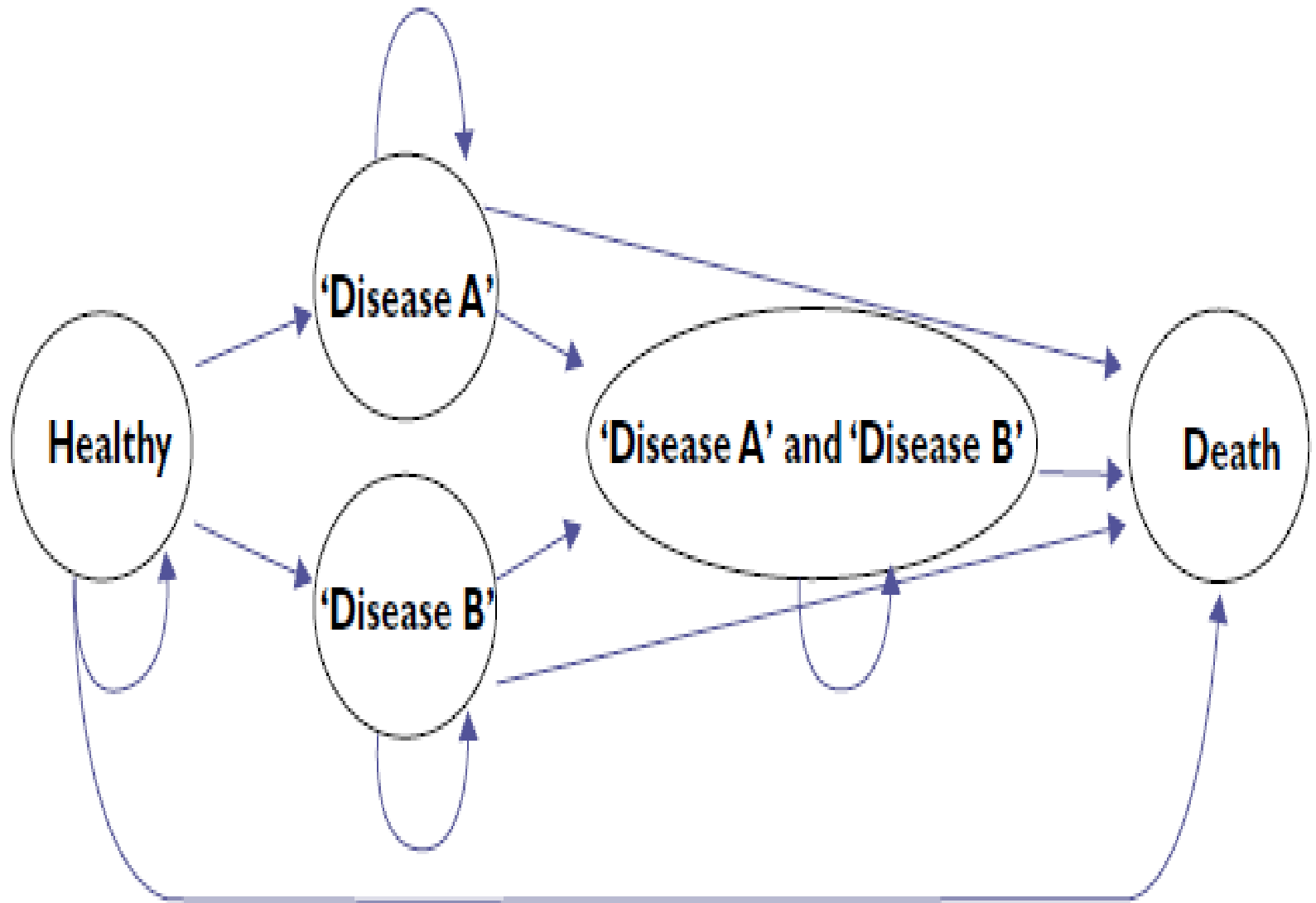
Markovian Assumption

- Probability of transition depends only on **current health state residence** and not on past health states (memory less property)

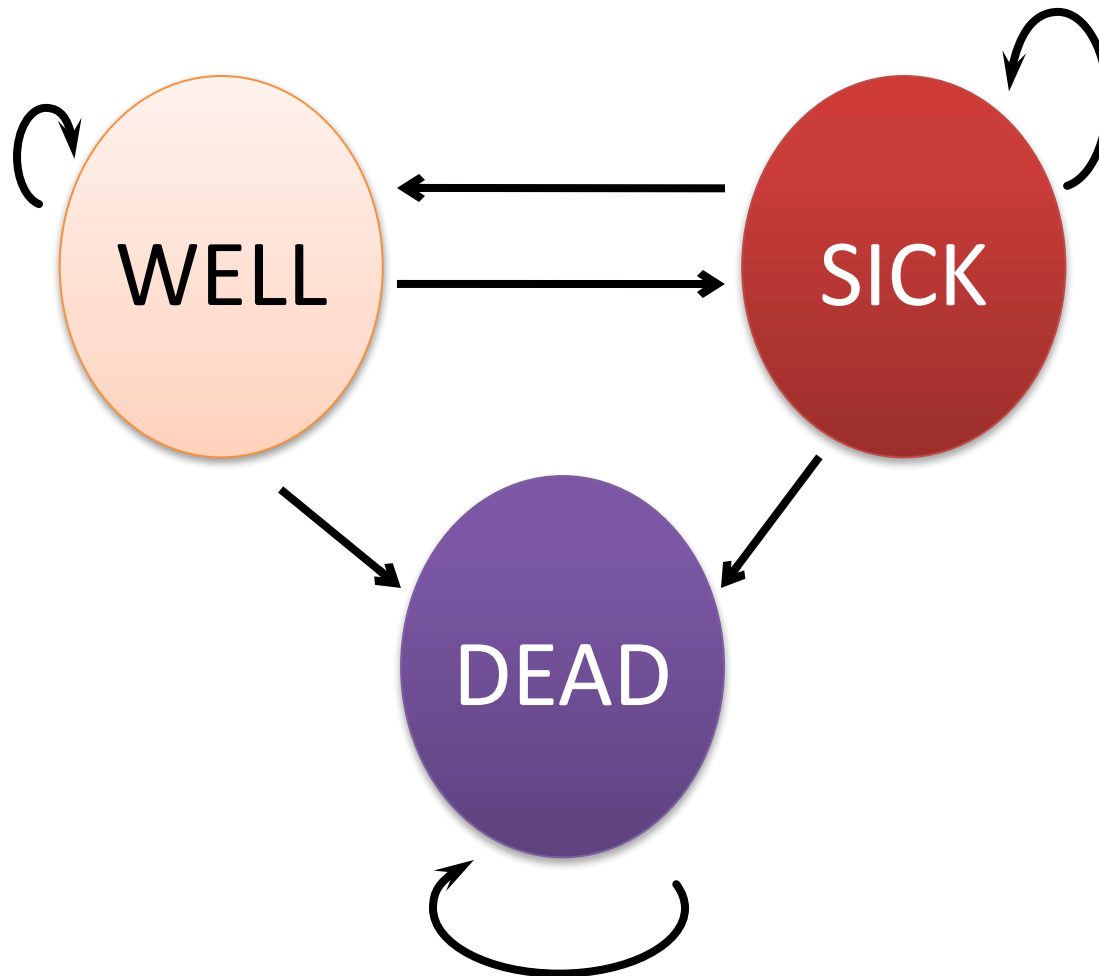
- Figure 3 illustrates an extension of the situation depicted in Figures 1 and 2, in which multiple sequential events over multiple time periods can be considered. After a person develops ‘disease A’, that person may later develop ‘disease B’, and *vice versa*.
- ***Recovery from*** either disease is **not assumed** to be possible (and hence there is no transition back to ‘healthy’).

- To capture all possible events, **multiple health states** need to be modelled, each with its **own set of possible transitions**. The exception is **death**, from which no transitions can be made.
- Death is considered **an absorbing state**.

- Analysis of **Markov decision** analysis trees does **not involve just a single transition** of subjects from left to right across the branches, but. rather involves repeated **recycling** b/n the terminal nodes (except in the case of absorbing states)
- A **Markov chain** describes a sequence of events in which the likelihood of an event occurring is dependent on the **preceding event**

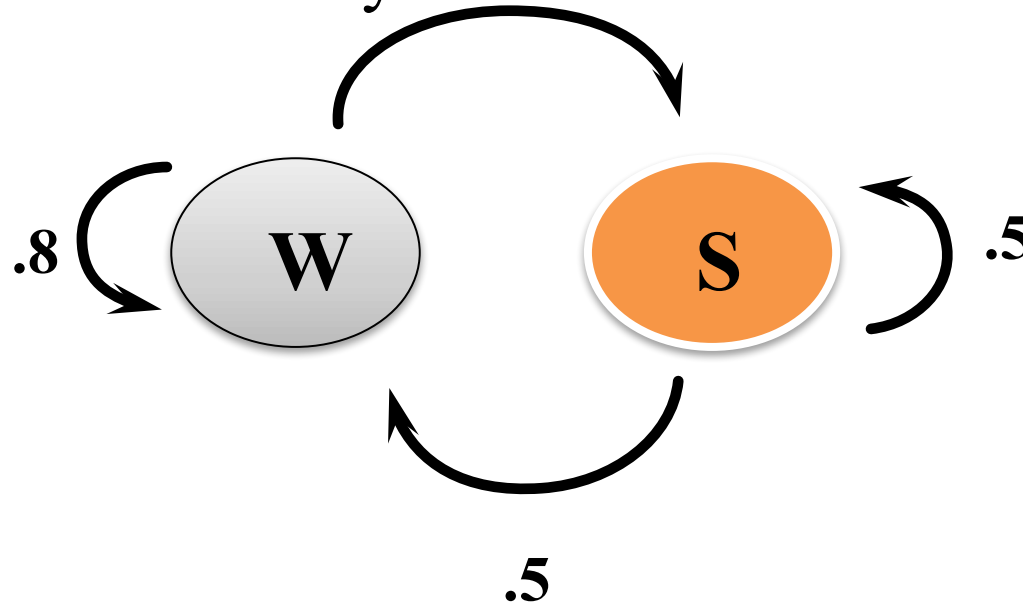


State Transition Diagram



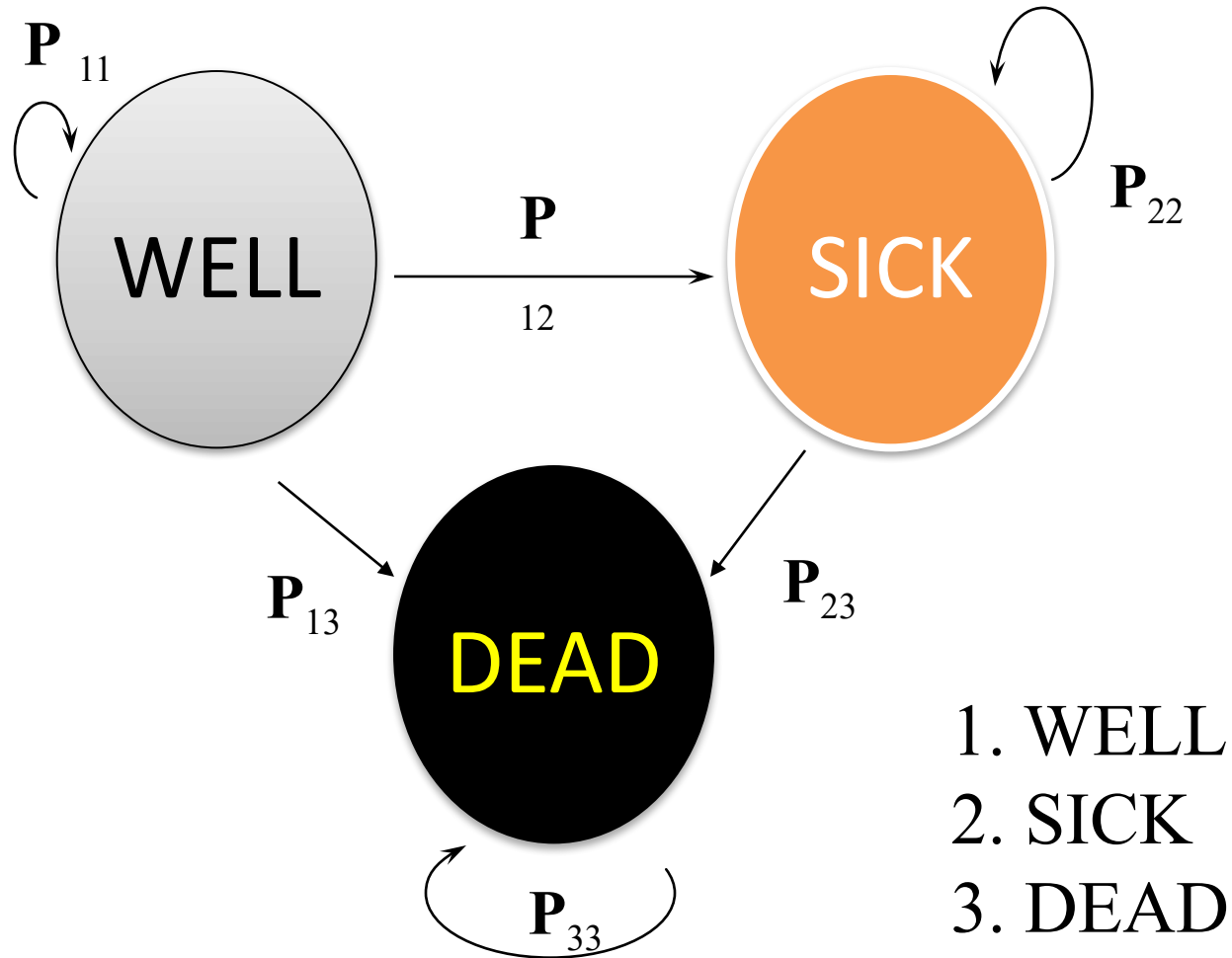
Methods . . .

Transition Probability- the chance that patients in a particular health state will transfer to another health state during the course of a cycle?



State Transition Diagram

P_{ij} = transition prob from state i to state j

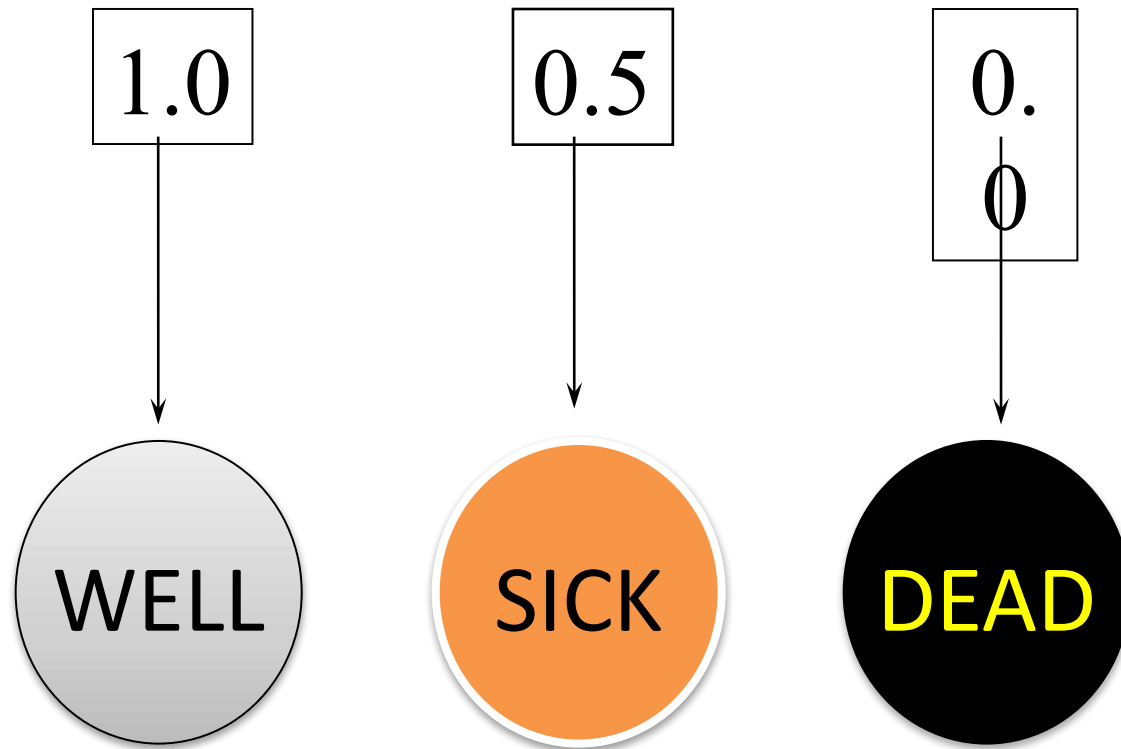


Methods . . .

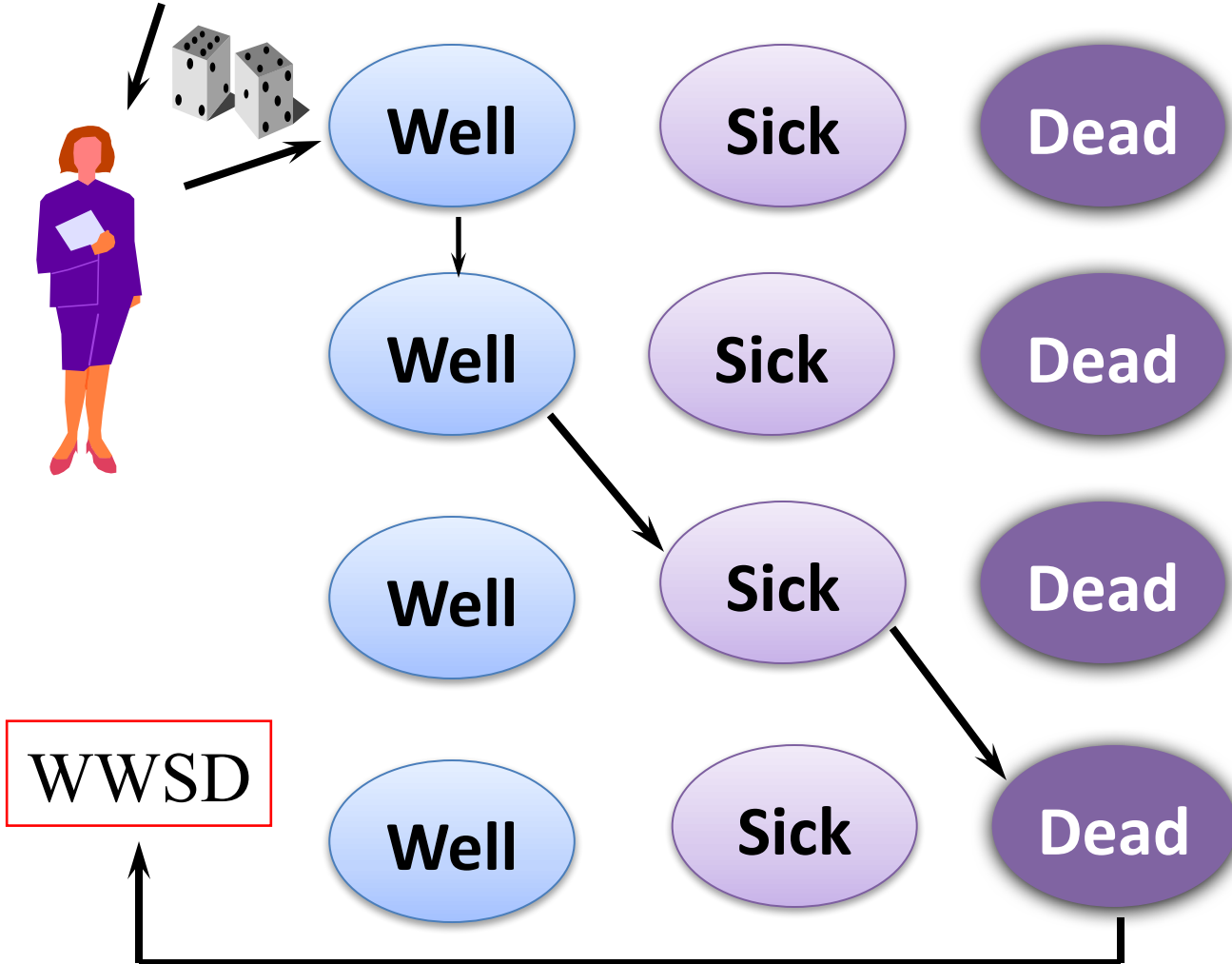
Transition probability matrix:
Another useful tool

P1	NEXT PERIOD	
	Well	Sick
THIS PERIOD		

Quality-of-Life Adjustments

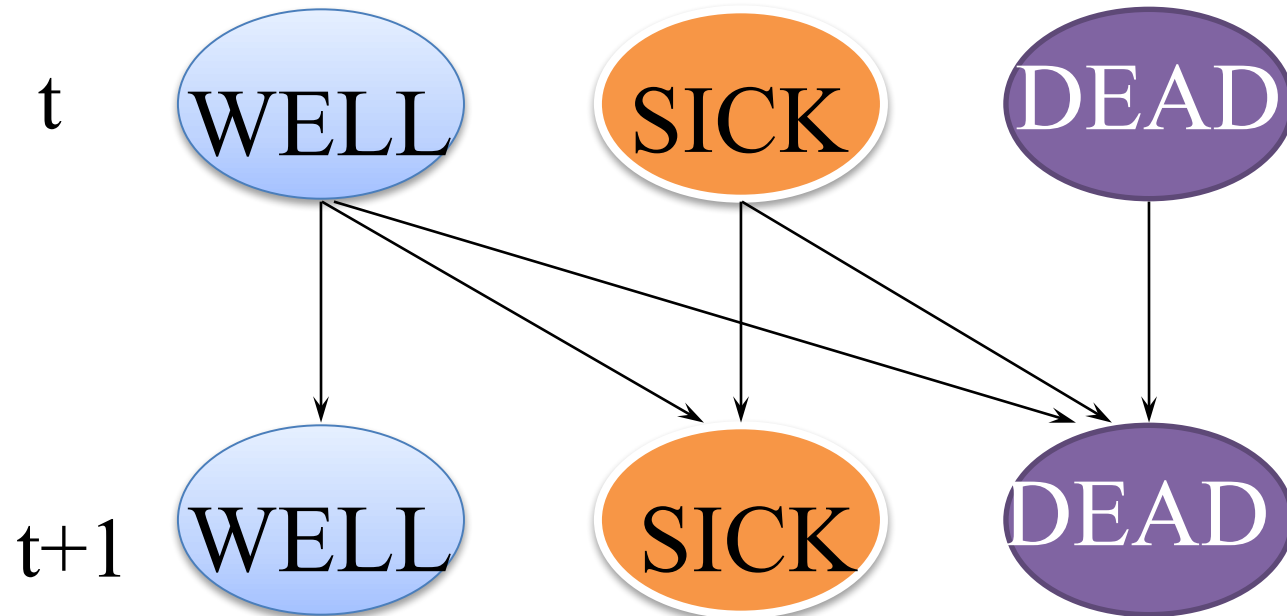


Monte Carlo Simulation

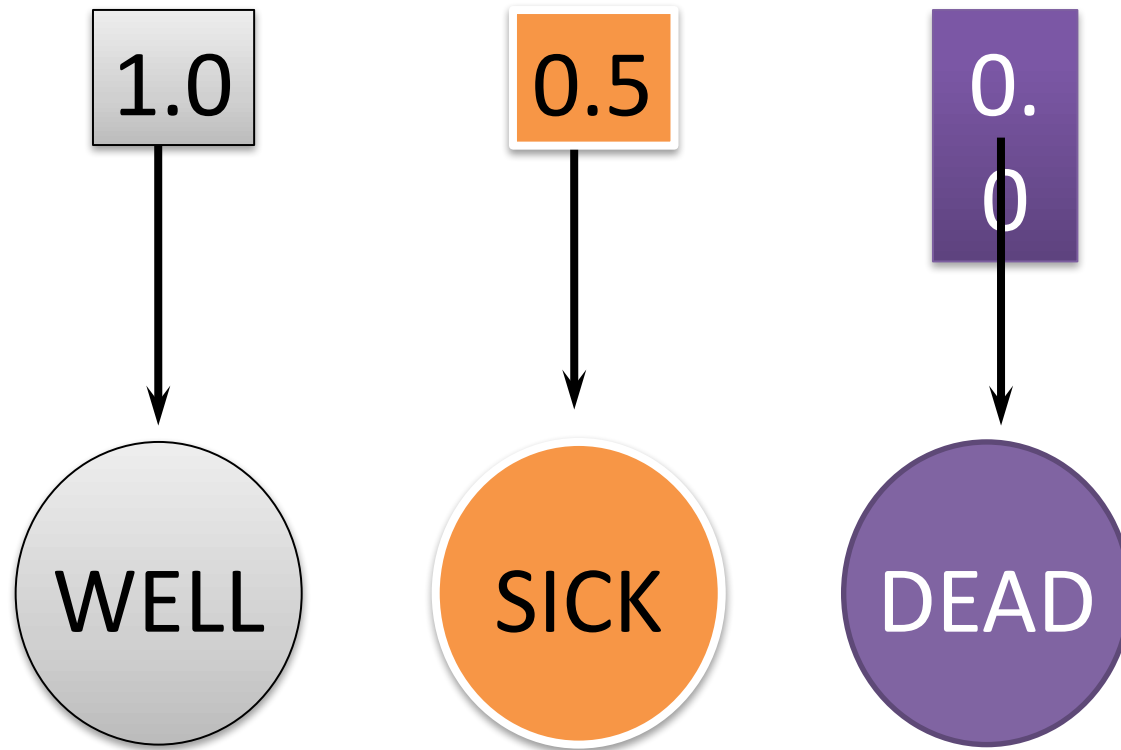


Cohort Simulation can Estimate ~~Survival Curves~~

Time



Quality-of-Life Adjustments



Transition Probability Matrix:

	WELL	SICK	DEAD
WELL	0.75	0.20	0.05
SICK	0	0.70	0.30
DEAD	0	0	1

Suppose everyone starts in the WELL state

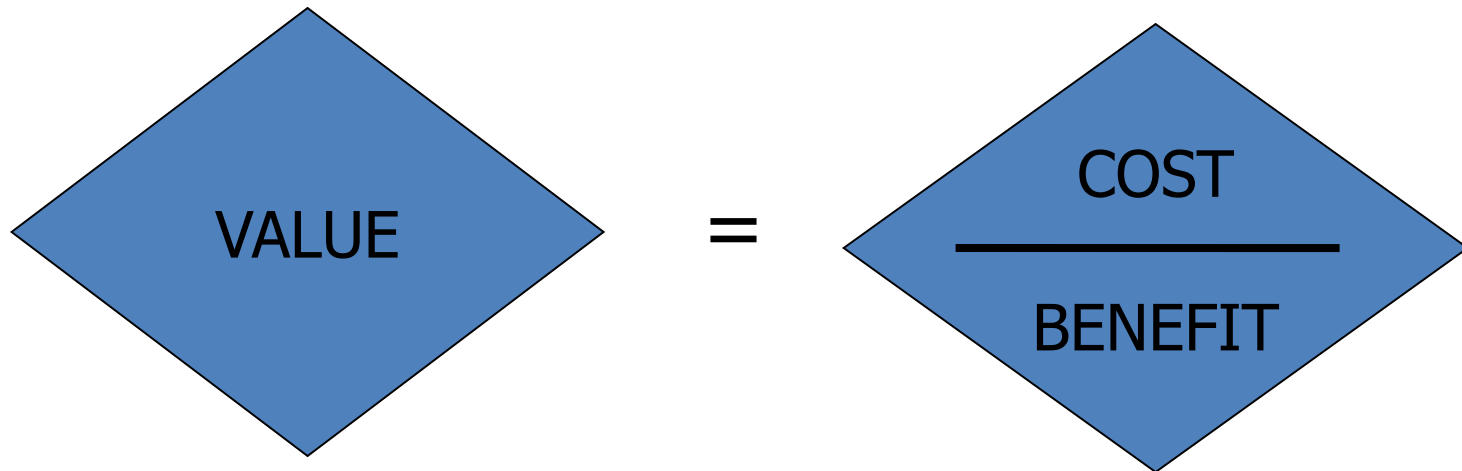
Main advantage of incorporating Markov modeling

- It allows for the simulation of **more complex consequences of an option**
- Not only can a greater number of possible events be simulated, they can be simulated for **lengthier periods** downstream.

limitation of Markov models

- with any cycle, the possible transitions which individuals can make depend **only on the state they are in**, and not on **how they arrived** at that state.
 - That is, Markov modeling lacks memory.

VALUES in health care system



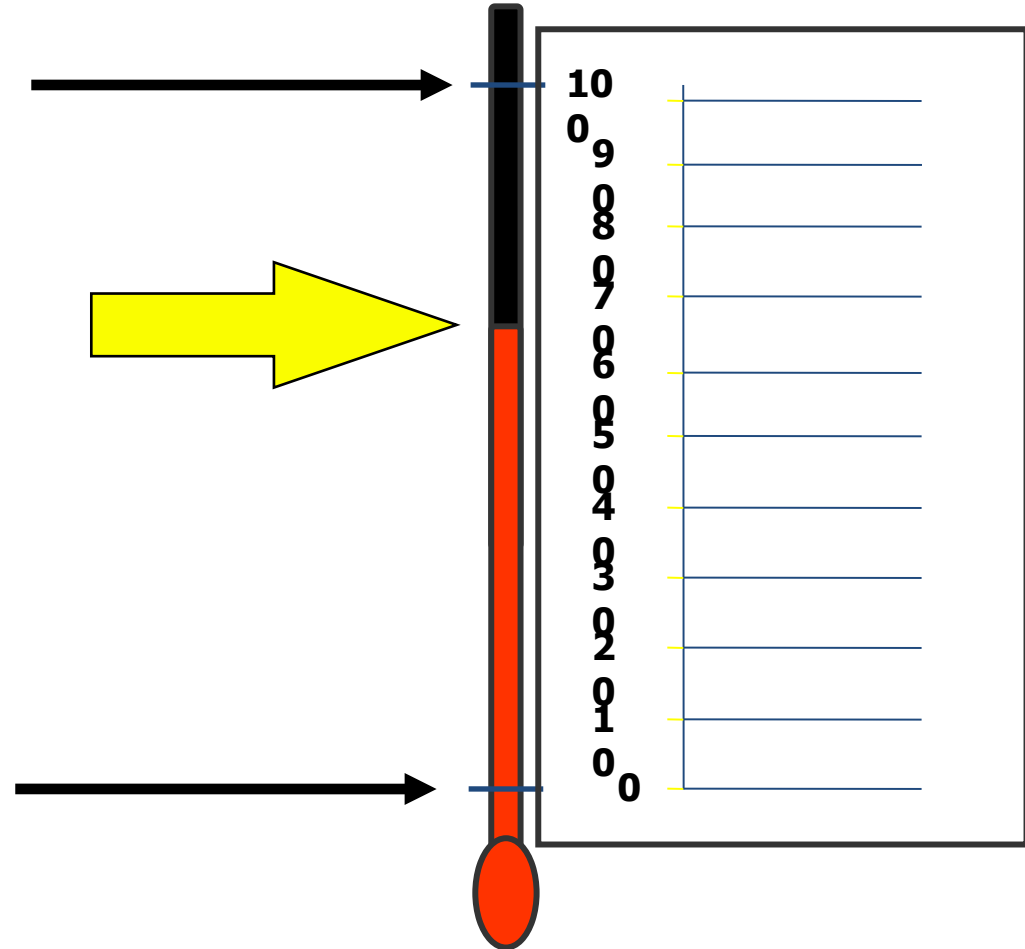
How to measure quality of life?

„thermometer of quality of life “

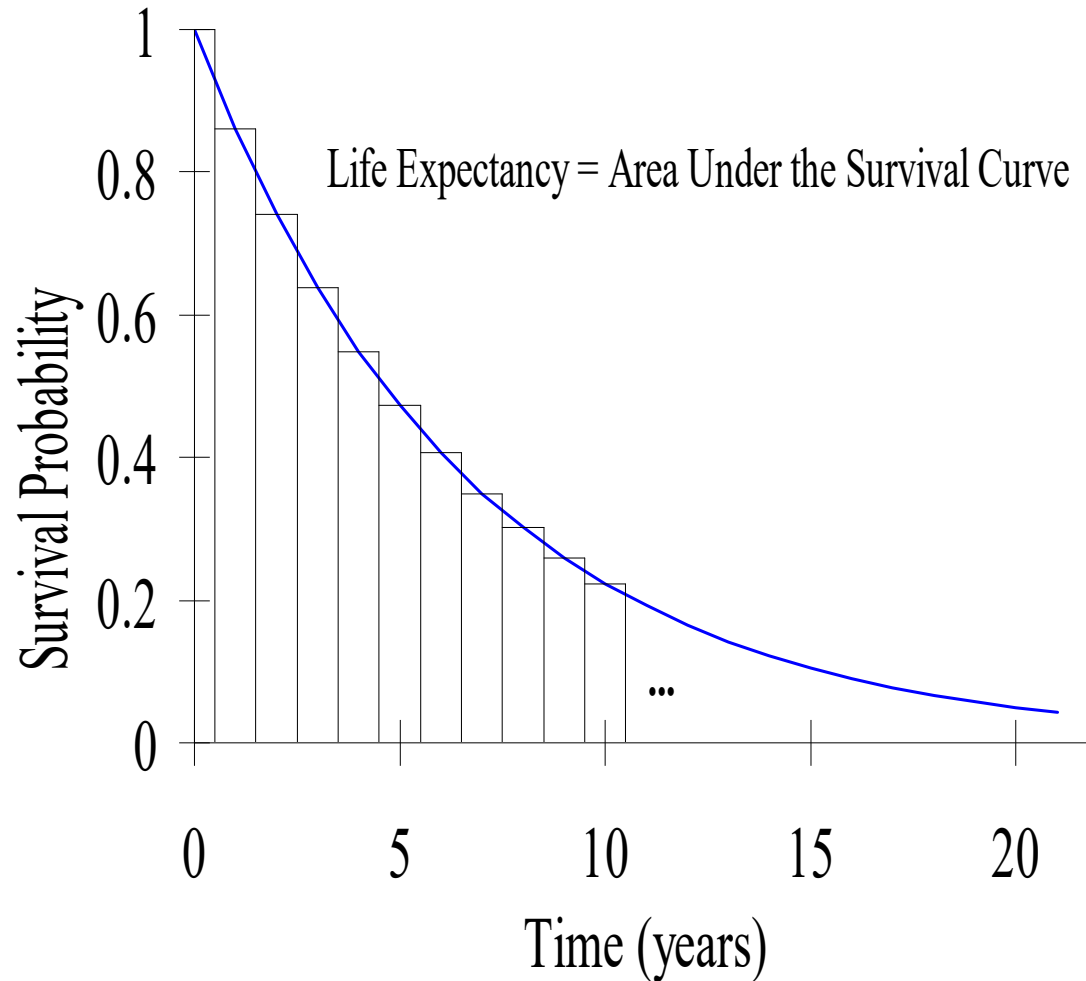
Maximal possible
health

choice of
patient
e.g. 0.65

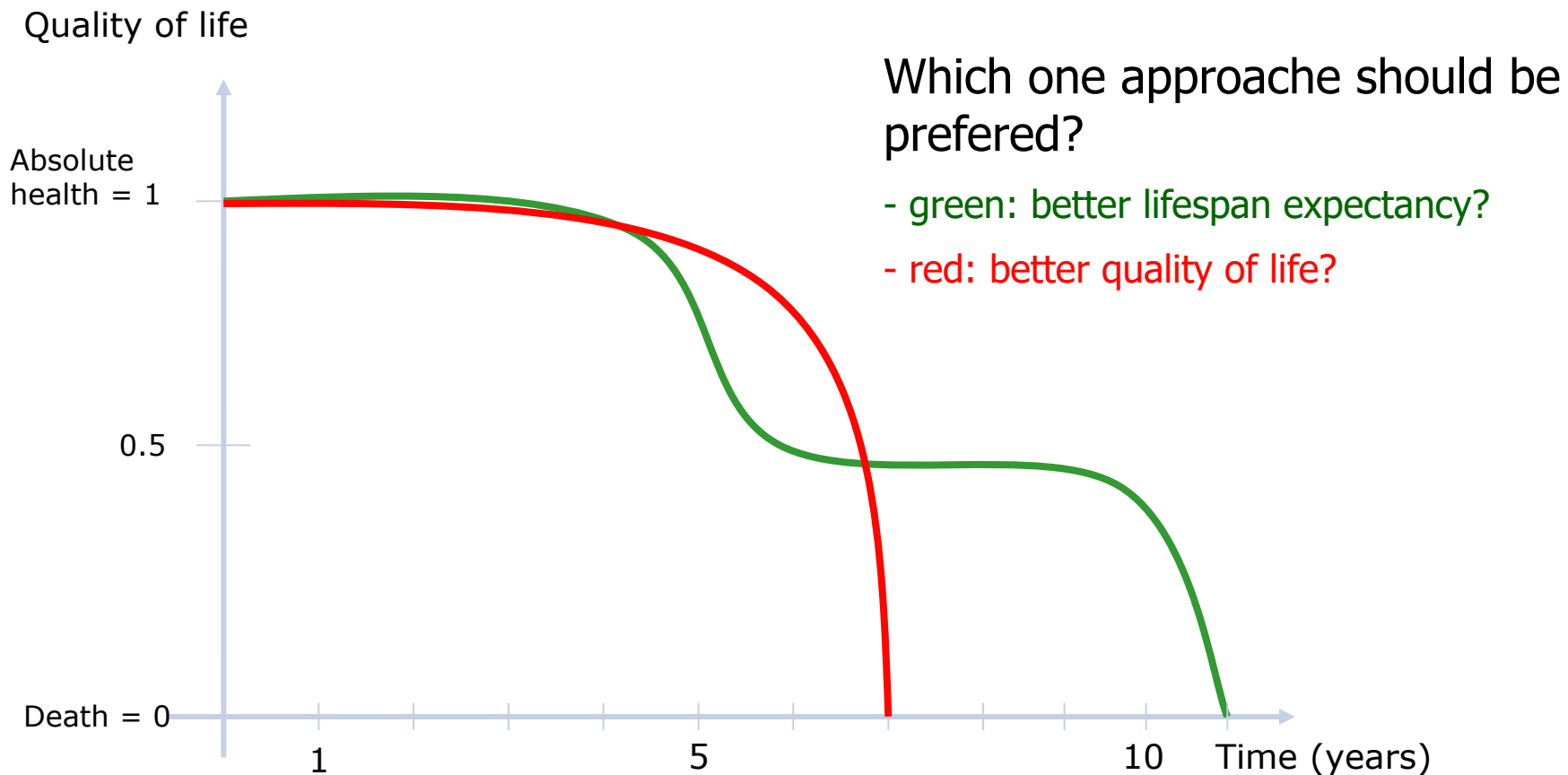
Death



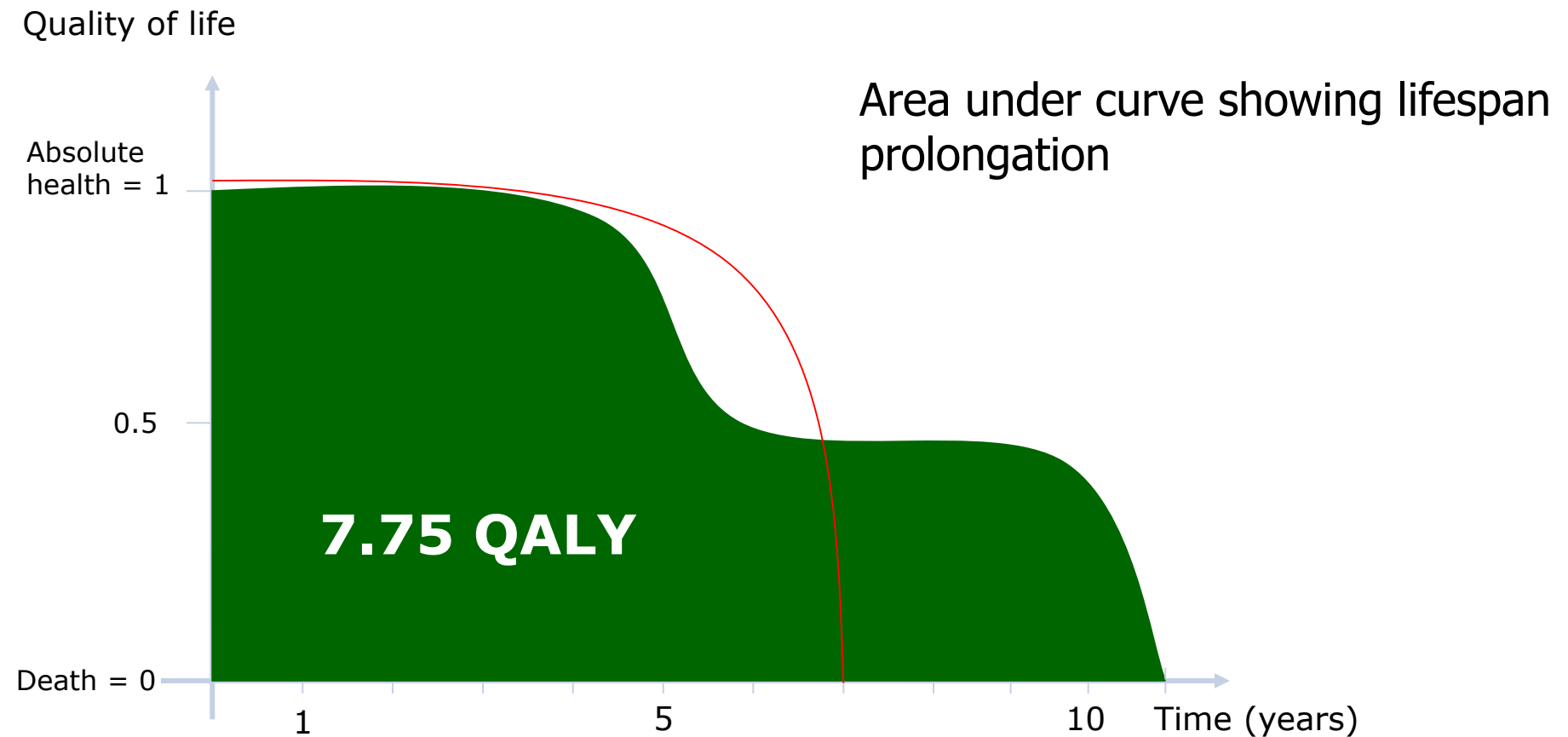
Markov Approximation to LE



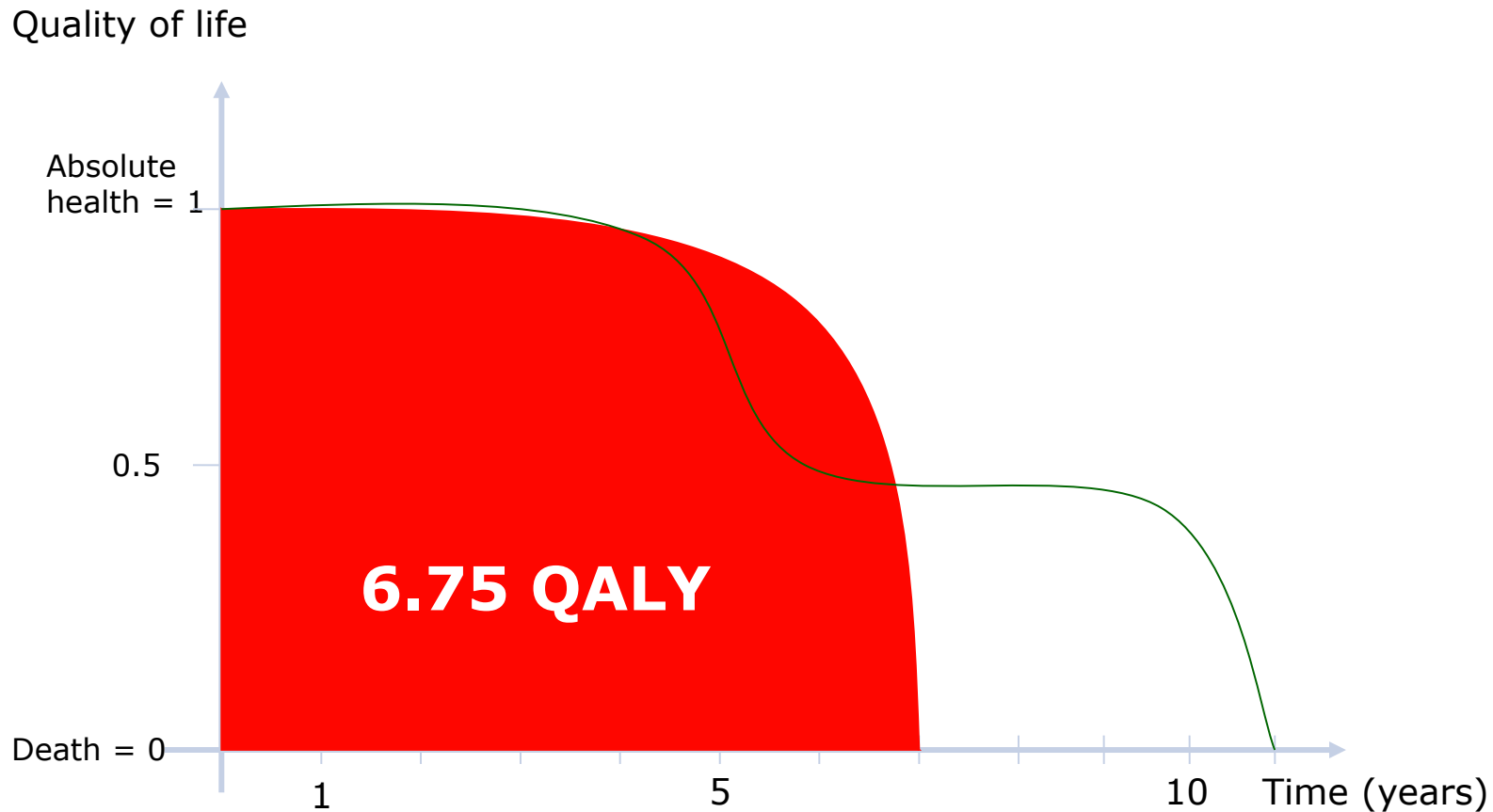
Two approaches with distinct lifespan expectancy and quality of life



QALYs – green curve



QALYs – red curve



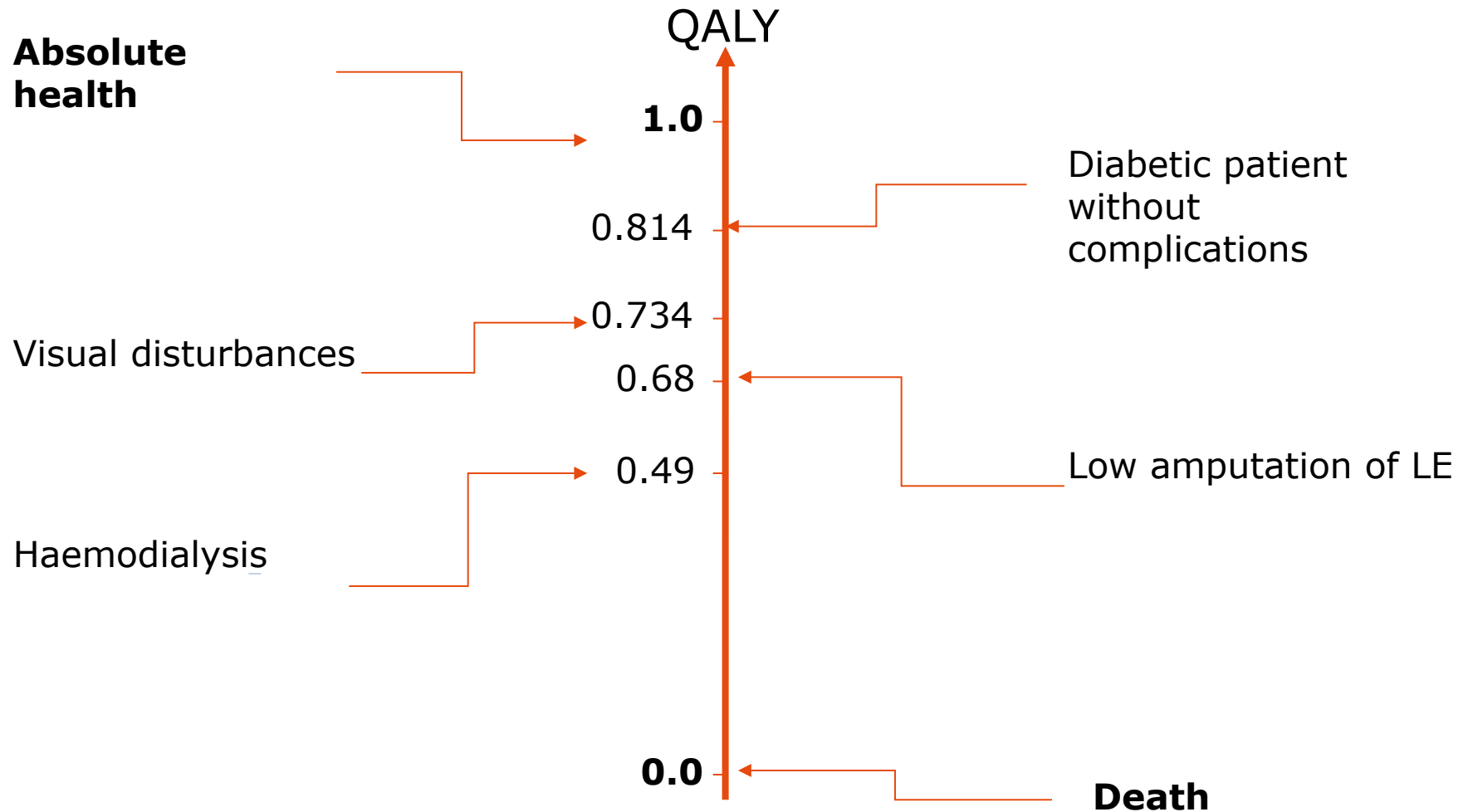
Comparison of previous two approaches

- In this case **green** option is more suitable because lifespan prolongations means higher value of QALY than in **red** option:

$$7.75 \text{ QALY} > 6.75 \text{ QALY}$$

Crux: Is this compatible with patients' attitude? ...e.g. diabetic patient has to comply with dietary restrictions etc...

QALYs in diabetic patients



Calculate Utilities

- Selected utilities from rating scale
 - 1.0 Completely healthy
 - .84 Kidney transplant
 - .58 Hosp. dialysis (pts)
 - 0.0 Dead
- 1. if dialysis extends a life 10 years at .58 then QALYs is
$$U * \text{years gained} = 5.8 \text{ QALYs}$$

Based on CEA (no adjustment for quality)
which option would you pick?

1. Option A
2. Option B
3. Need ICER

Option	Cost	YLS	QALYS
A	R\$5000	6 years	$0.8 * 6 =$ 4.8 QALYS
B	R\$4000	10 years	$0.3 * 10\text{LYS}$ 3.0 QALYS

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which option would you pick?

1. Option A
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Calculating and Interpreting Cost-Effectiveness Ratios

1. Consider the following information about the costs and effects of four alternatives for treating a given health problem. Which alternative should be funded and why?

<u>Alternative</u>	<u>Total Cost</u>	<u>Total No. of Lives Saved</u>
A	\$300,000	30
B	\$300,000	36
C	\$500,000	40
D	\$600,000	45

CHAPTER FOUR

Costs and Time Preference

4.1 Discounting of Costs

The time value of money

- There is a **time value** associated with money.
- We say that money has a time value because that money can be invested with the expectation of earning a positive rate of return.
- There fore, **money received today is worth more than the same amount of money received next year.**
- In other words, “a dollar received today is worth more than a dollar to be received tomorrow”.
- That is b/c today’s dollar can be invested so that we have $>$ one dollar tomorrow

Justification for the “present value” approach;

- Inflation
- Technological innovation
- Empirical evidence of positive time preference

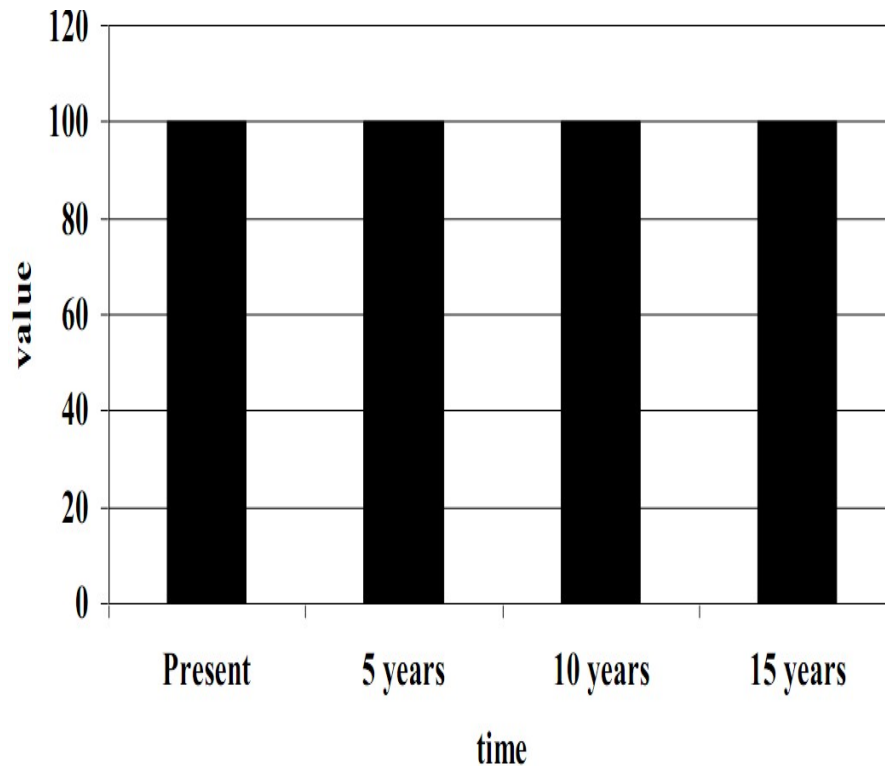
Would you prefer to have \$100 now or in the future?

- Positive rate of time preference
- Why?

Reasons for discounting future costs

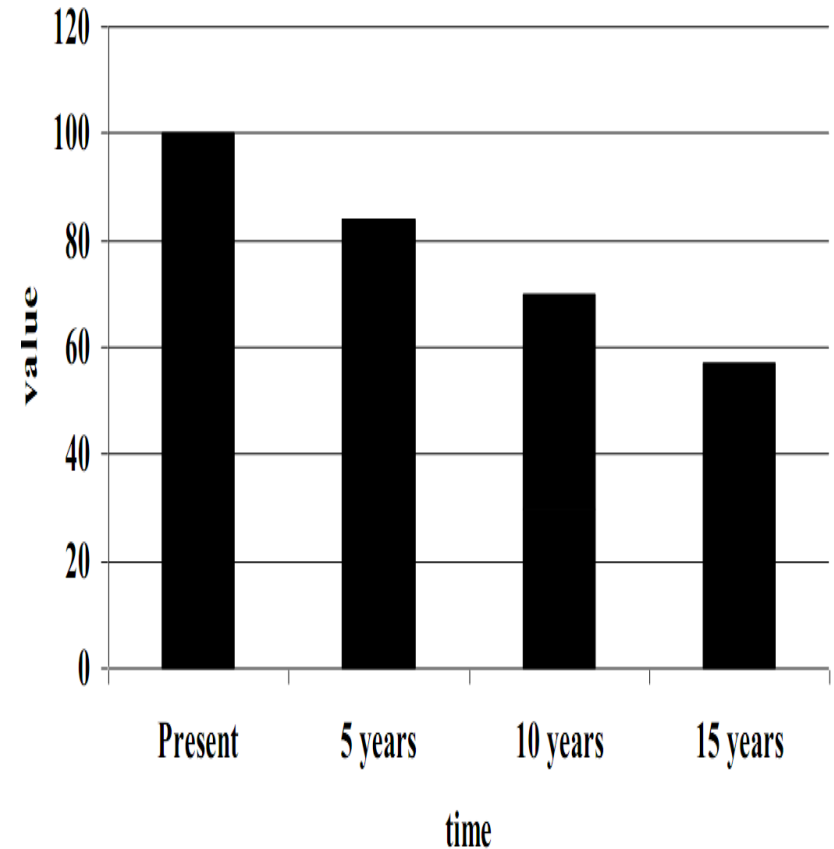
- Human beings prefer to put off payment of **costs** into the **future**.
- Therefore a cost incurred in the future seems/is of less importance than the same cost incurred in the present.
- Therefore future costs must be adjusted to reflect present value (e.g. a \$1000 cost one year from now requires only \$930.00 in hand today assuming a 7% return on investment.

...Discounting....



Assuming zero inflation and zero bank interest

Would you prefer to have
\$100 now or in the future?



- In order to reflect this **lesser value** of a deferred or future cost, we **discount** the **future cost** to a “**present value**”, e.g., \$1000 becomes \$950 when viewed from the present.
- Discounting which assumes that people prefer **benefits** which occur **sooner** rather than later.
- Costs and consequences are discounted to present values by use of a **discount rate**.

- Discounting adjusts future costs or benefits using an expected interest or *discount rate*.

Present value of a stream of costs over time

$$= \sum C_t / (1 + r)^t$$

$$= C_1 / (1 + r)^1 + C_2 / (1 + r)^2 + C_3 / (1 + r)^3 + \dots$$

Where C_t = costs at time t and

r = discount rate (typically ranges from .03 to .06) and

t = the number of years in the future.

Discounting example

- You wish to implement a diabetes management program which will cost you \$1500 per year.
 - The benefits from this program won't be evident for 2 years, so you want to evaluate it after 4 years.
- Use $r = .05$

Discounting example 1

Year	Costs	PV
1	1,500	1,429
2	1,500	1,361
3	1,500	1,296
4	1,500	1,234
total	\$6,000	\$5,320

Discounting example 2

Consider a cost of \$1000 incurred at end of years 1, 2, 3, & 25, where $r = 5\%$

Year	Future Cost	Present Value
1	\$1000	\$952
2	\$1000	\$907
3	\$1000	\$864
25	\$1000	\$295
Total Costs	\$4000	\$3018

...Discounting....

The terminology of time value

- **Present Value** - An amount of money today, or the *current value of a future cash flow*.
- **Future Value** - An amount of money at some future time period.
- **Period** - A length of time (often a year, but can be a month, week, day, hour, etc.).
- **Interest Rate** - The compensation paid to a lender (or saver) for the use of funds expressed as a percentage for a period (normally expressed as an annual rate).

Abbreviations:

- PV - Present value
- FV - Future value
- Pmt - Per period payment amount
- N - Either the total number of cash flows or the number of a specific period
- i - The interest rate per period

Discounting...

- The present value of cost c in t years at interest (also known as “discount”) rate i is:

$$PV = \frac{c}{(1+i)^t}$$

- How much would you pay today for a promise of receiving \$100 in five years?
- $PV = 100 / (1+i)^5$
 - If we use an interest rate of 7%, we get:
 - $PV = 100 / (1.07)^5 = \$71$
- *Note that without discounting ($i=0$), the PV would be \$100.*

Discounting...

- The future value of \$ c in t years at interest rate i is: $FV = c(1+i)^t$
- Suppose you invest \$100 today. How much is it worth next year?
- $FV = 100 * (1+i)^1$; *interest rate of 7%,*
- If we use an interest rate of 7%, we get:
- $FV = 100 * (1.07)^1 = \$107$
- *FV in two years would be $100 * (1.07)^2 = \$114.50$*
- *Note that without interest ($i=0$), the FV would be \$100*

Discounting...

Example:

- Suppose you place \$100 in the bank immediately, another \$100 one year from now, \$500 two years from now, and \$500 three years from now. How much will you have in the bank in 5 years? (use an interest rate of 5%)

Answer:

- $$FV = 100*(1+i)^5 + 100*(1+i)^4 + 500*(1+i)^3 + 500*(1+i)^2$$

If we use an interest rate of 5%, we get:

- $$FV = 128 + 122 + 579 + 551 = \$1,379$$

Note that without interest ($i=0$), the FV would be \$1,200.

Pharmacoeconomics assignment

- Types of Intervention (Cost Studies Collection and Types of Costs)
- Adjusting for Inflation and Annuitizing Capital Expenditures
(rating scale, standard gamble, and time tradeoff)
- Rating Methods used to measure and estimate utility rating scale
- **standard gamble** Methods used to measure and estimate utility
- **time tradeoff** methods used to measure and estimate utility

CHAPTER FIVE

Cost Benefit Analysis

Economic evaluation

- Full economic evaluations include **cost-minimization, cost-benefit, cost-effectiveness, and cost-utility analyses.**
- Each method is used to compare competing programs or treatment alternatives.
- The methods are all **similar in the way they measure costs (in dollars) and different in their measurement of outcomes.**

Economic evaluation...

- **Although a full economic evaluation generally provides higher-quality and more useful information; the time, resources, and effort employed are also great**
 - Thus healthcare practitioners and clinicians also find it necessary to employ various partial economic evaluations.

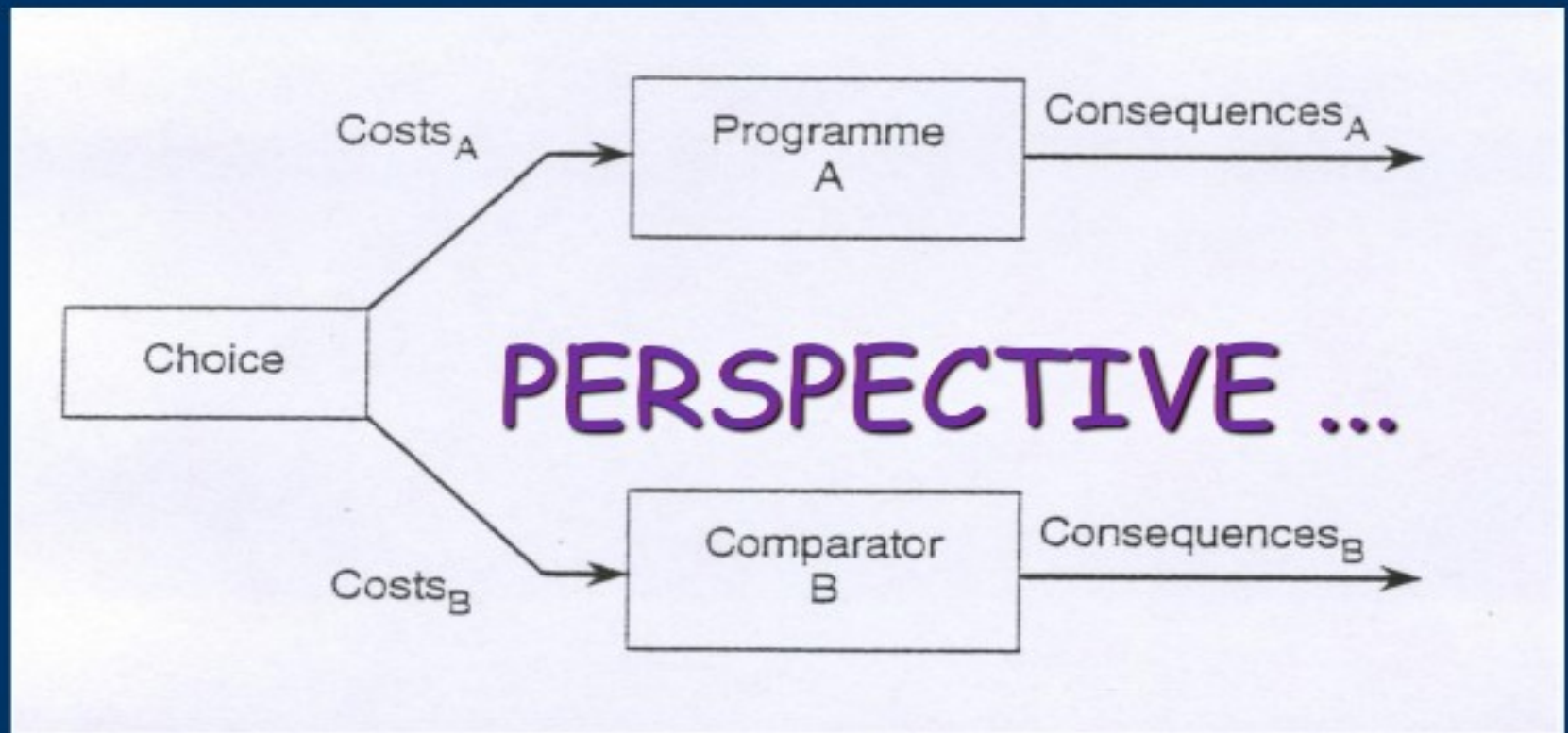
Definition: Economic Evaluation - is the identification, measure, and comparison of the costs (i.e. resources consumed) and outcomes (clinical, economic, and humanistic) of interventions (pharmaceuticals, non-drug therapies, public health programs).

Economic evaluation

- Purpose of economic evaluation - efficient resource allocation
- In economic evaluation, the questions that should be answered
 - Is there a comparison of two or more alternatives?
 - Are both costs and consequences examined?
- If not-the study is not an economic evaluation but may be:
 - description of costs or outcomes
 - evaluation of efficacy or effectiveness
 - cost analysis

Economic evaluation

Economic evaluation always involves a comparative analysis of alternative courses of action



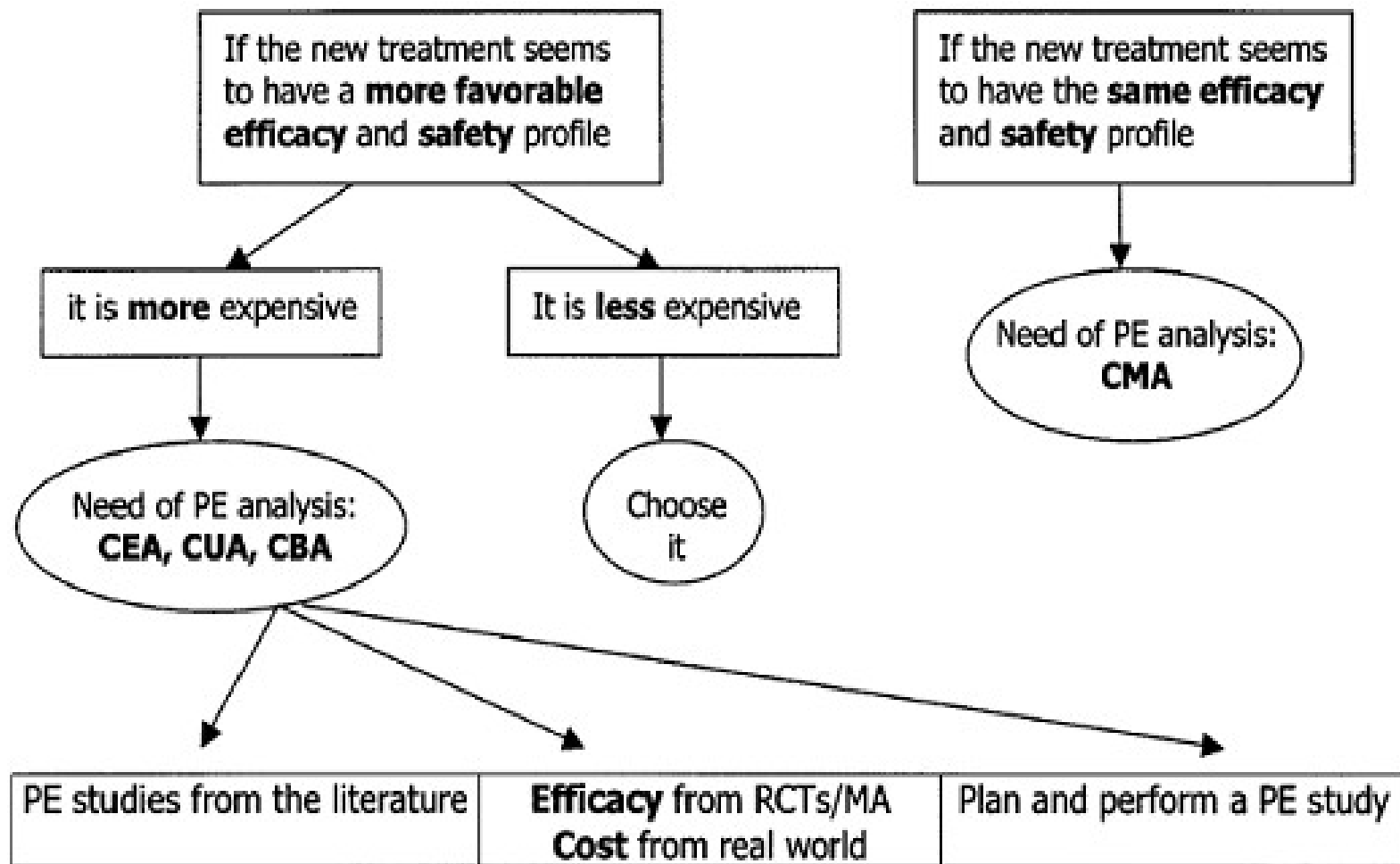
Economic evaluation

Who uses Economic Evaluation?

- Managers in hospital or health care plans
(e.g. formulary decision)
- Pharmaceutical companies
- Government /Policy makers
- Researchers, etc.

Economic evaluation

Figure 1 Process for Incorporating Pharmacoeconomic Evaluations in Formulary Decisions



Legend: RCTs = (randomized controlled trials); MA = meta-analysis

Economic evaluation

Types of analysis in Economic Evaluation

- Cost Minimization Analysis(CMA)
- Cost Benefit Analysis(CBA)
- Cost Effectiveness Analysis(CEA)

Economic evaluation...

- Cost Utility Analysis (CUA)
- Cost-of-Illness Analysis (COI)
- Cost Consequence analysis
- Cost-offset Analysis
- Budget Impact Analysis (BIA)

5. Cost benefit analysis (CBA)

- CBA is an evaluation method for **comparing the monetary value of all resources consumed (costs)** in providing a program or intervention with the **monetary value of the outcome (benefit)** from that program or intervention.
 - CBA seeks to place monetary values on both the inputs (costs) and outcomes (benefits) of health care.

- Evaluates the value of all resources consumed in implementing a program or intervention against the value of the outcome (Clinical, Economic, Humanistic) in terms of dollars.
- The aim is to take **a societal perspective**

Advantages over other economic evaluation

- CBA allows uniform comparison of programs or interventions with **entirely different outcomes** (e.g. cost benefit of having a neonatal care program vs a cardiac rehabilitation program)
- **It** Allows comparison of health program with non-health program

Advantages over other economic evaluation

- Most comprehensive and theoretically sound form of economic evaluation(*but in practice the most difficult and most criticized*).
- Can readily combine health and non health outcome associated with a single program

Disadvantage :

- CBA is **difficult to perform** because it requires both cost and benefits to be measured in monetary terms
- Many benefits are either **difficult to measure, difficult to convert to dollars or both** (patient satisfaction with the health care system)
- CBA ignore **many intangible but very important benefits not measurable in money terms**, e.g. relief of anxiety.

Disadvantage ...

- CBA may also seem to **discriminate against** those in whom a return to productive employment is unlikely, e.g the elderly, or the unemployed.
- Productivity and quality of life is difficult to perform reliably and meaningfully

CBA...

Problems /Disadvantage with CBA

- difficult to give monetary values to health outcomes.
- Result depends on dollar values assigned to life

When to use CBA:

- When a policy maker has **a broader perspective** and is faced with one or more of the following,
 - Must decide whether to implement a specific program.
 - Required to choose among competing options.
 - Has a set budget and must choose and set priorities from a group of potential projects.
 - The intervention under consideration could produce a number of widely differing outcomes.

CBA...

CBA can also be used:

- CBA should be employed when comparing treatment alternatives in which **the costs and benefits do not occur simultaneously.**
- when comparing programs with **different objectives because all benefits are converted into dollars.**
- to evaluate a single program or compare multiple programs.

CBA can also be used...

- However, valuing health benefits **in monetary terms can be difficult and controversial.**
- The expression of some health **benefits as monetary units is neither appropriate nor widely accepted.**
- Therefore, unless the benefits of a program or treatment alternative are expressed appropriately in dollars, CBA should not be employed

CBA...

- If the interventions result in a stream of benefits and costs over time --> Choose a discount rate and construct present value. That means *Future costs and benefits are discounted or reduced to their current value.*
- Both the costs and the benefits are measured and converted into equivalent dollars *in the year in which they will occur.*
- The costs and benefits are expressed as a ratio (a **benefit-to-cost ratio**), a net benefit, or a net cost.

- A clinical decision maker would choose the program or treatment alternative with **the highest net benefit or the greatest benefit-to-cost (B:C) ratio.**

Components of CBA

- **Two categories of costs; direct medical and direct non medical,**
- **Three categories of benefits; direct benefits (both medical and non medical), indirect benefits (productivity), intangible benefits.**

CBA...

Guidelines for the interpretation of this ratio are indicated as:

- If the **B:C ratio** is > 1 , the program or treatment is of value- that is the benefits realized by the program or treatment alternative outweigh the cost of providing it.
- If the **B:C ratio equals 1**, the **benefits equal the cost-** that is the benefits realized by the program or treatment alternative are equivalent to the cost of providing it.



Guidelines for the interpretation of this ratio...

- If the B:C ratio is < 1 , the program or treatment is not economically beneficial- that means the cost of providing the program or treatment alternative outweighs the benefits realized by it.

CBA...

- Of all pharmacoeconomic evaluation methods, CBA is **probably used the least.**
- Although this method has the advantage of valuing indirect costs monetarily and intangible benefits (using the WTP approach),
 - the valuation of outcomes such as productivity and quality of life is difficult to perform reliably and meaningfully.
- Because of difficulties in measuring indirect and intangible benefits, many **CBAs measure and quantify direct costs and direct benefits only..**

- Therefore, CBA is difficult to perform because it requires that both costs and benefits be measured in (or converted into) monetary terms.

- i. Human Capital Approach.

- ii. Willingness-to-Pay Approach.

Expected Value

- When values of costs or benefits are not known with certainty, but are known with probability, then expected values are used.
- Expected value of a benefit is: $E(B) = \text{Prob}(B=b_i) \times b_i$
 - where: $\text{prob}(B=b_i)$ is the probability that the benefit is worth \$ b_i .

CBA...

Expected Value.....

Example: Mauskopf, J.A. et. al, “Economic Impact of Treatment of HIV-Positive Pregnant Women and their Newborns with Zidovudine: Implications for HIV Screening,” JAMA 276: 2, 132-8, July 10, 1996.

- Probability of maternal-to-fetal transmission when the mother is HIV-positive
 - No Treatment = 25.5%
 - With Zidovudine Treatment = 8.3%
- Cost of Zidovudine treatment = \$1,045
- Lifetime cost of treatment of an infected child from birth = \$98,915

CBA...

- Expected value of cost of a lifetime pediatric HIV infection
= probability of transmission x lifetime treatment costs
 - No Treatment = $.255 \times 98,915 = \$25,223$
 - With Treatment = $.083 \times 98,915 = \$8,210$
- Expected benefits of treatment = Expected costs averted by treatment
= $25,223 - 8,210 = \$17,013$
- Cost of Zidovudine treatment = \$1,045
- Expected Net Benefits = $17,013 - 1,045$
= \$15,968 per HIV-positive pregnant woman

CBA...

Steps in Conducting CBA

- Define/determine the type of program or intervention to be considered.
- Identify alternatives
- Identify the effects of the intervention for all health and non health outcomes and classify them as either benefit or costs.
- Assign values
- Determine and calculate summary measures
- Evaluate the results with SA.
- Prepare results.

- Various methods have been developed to estimate the monetary value of health benefits. The two most common methods seen in the pharmacoeconomic literature are the **HC approach** and the **WTP approach**, which are going to be discussed in this class.

CBA...

I. **Human capital approach (HCA)**

- For years the most commonly applied method of benefit measure was the HC approach.
- The HCA means that **the value of people's contributions is linked to what they are paid.**
- Value of health benefits=the economic productivity they permit
 - Cost of disease=lost productivity
 - Cost of a sick day=how much you earn that day

HCA...

- Value of life = present value of lifetime earnings (= lifetime productivity in competition)
 - represents productivity gains from extending life (Benefit side) or
 - Productivity loss due to early death (cost side)
 - for society as a whole, represents a loss in national output due to mortality.

CBA...

- The HC approach assumes that the value of health benefits equals the **economic productivity that they permit.**
- There are two basic components to calculating HC: **wage rate and missed time (days or years) because of illness.**

Wage Rate calculations

- Depending on the type of study, a yearly wage rate or a daily wage rate can be calculated.

CBA...

- **A yearly wage rate (income per year) would be calculated for a program or intervention that would reduce long-term disability or death.**
- For example, a pneumococcal vaccination program might result in preventing premature death. Thus, it would be appropriate to use **a yearly wage rate** and assess the value of **the number of years saved because of the intervention.**

CBA...

- **A daily wage rate** (income per year divided by number of days worked per year) may be calculated for a program or intervention targeted at an acute or chronic illness **with short-term disability.**
- A person may not be adversely affected by the disease state on a continual basis, but he or she may **have short-term periodic disability.**
 - For example, asthma, a chronic disease state, may include episodic asthma attacks. Thus, **a person may only experience problems with the disease state on a periodic basis.**

- For this type of disease state, a daily wage rate would be calculated.

To calculate **a daily wage rate both income and number of days worked per year must be assessed.**

We may assume that the average person works 240 days a year when accounting for weekends, vacation, and sick leave.

<i>Categories</i>	<i>Examples</i>
Missed work	Days missed from work (for employed)
Missed housekeeping	Days missed from housekeeping (for unemployed)
Restricted activity days	Percent of time during which work or housekeeping was restricted Did not miss an entire day of work or housekeeping but not productive for part of the day
Caregiver time	Parent's time spent as a caregiver to a child who has an illness

Using the asthma clinic example, we will calculate an indirect benefit.

- Assume that the population served by the clinic is made up of adults with an average income of \$40,000 and 240 days worked per year. The daily wage rate (average income/number of days worked per year) would be $\$40,000/240 = \$167/\text{day}$.

CBA...

- An average of 20 days a year were missed from work before participating in the asthma clinic, and an average of 7 days a year were missed from work after participating in the asthma clinic. Multiplying the daily wage rate times the number of missed days results in the value of lost productivity.
- In other words, the value of 20 days lost from work is \$3,340, and the value of 7 days lost from work is \$1,169.
 - The difference between before and after the program is **\$2,171**, which is the **cost savings or the indirect benefit of the program or intervention.**

Limitation of HCA

1. The method excludes **cost of pain and suffering due to illness** - It **ignores the non-financial costs of pain** and suffering, which are often associated with illness.
2. It **does not assign any cost of illness to persons who are retired or live off non-labour income.**
 - People who are **not working for pay** (e.g., homemakers, students, retirees) **are valued at 0!** (Even for the employed, time away from the job is valued at 0.)
 - valuing benefits in terms of rates of **pay neglects the health benefits that accrue to people who are not employed-for example, non-working wives and retired people.**

3. Wage difference may **reflect wage discrimination instead of differences in productivity.**

–Implies that **people with higher wages have higher social value.**

4. The **precise social rate of discounting is unknown.**

Although biases exist with this method, *it is the most commonly used method to measure indirect benefits.*

Advantages of CBA...

- *It is fairly straightforward and easy to measure.*
 - **Income estimates** can be obtained or estimated from **publicly available sources**, and **days lost from illness** can be readily obtained from the **patient or another secondary source**.
- *According to the human-capital approach, the value of life is determined by **the contribution the individual could make to the social product**.*
- *Its relatively easy application is outweighed by **serious economic and ethical shortcomings**.*

CBA...

II. Willingness to pay approach (WTP)

- WTP approach seeks to establish the value that people attach to **health care outcomes** by asking them **how much they would be prepared to pay to obtain the benefits or avoid the costs of illness.**
- Individuals are asked to state maximum WTP or willingness to accept (WTA) to ensure that the welfare gain occurs or to tolerate the welfare loss.
- In contrast to the HC approach, the WTP approach is based on the concept of **subjective utility.**

CBA...

- Example; A study was conducted (by Thompson) using **willingness to pay methods to value the effect of a hypothetical cure on their state of health in 247 patients with rheumatoid arthritis** who were already enrolled in a randomized controlled drug trial.
- The precise question that they were asked was: **“What percent of your families' (i.e. household) total monthly income would you be willing to pay on a regular basis for a complete cure for arthritis?”**
- The results indicated that the average responder was willing to pay 22% of their household income to secure a cure for arthritis.

Source: Am J Public Health 1986;76:392-6.

Advantages of WTP as outcome measure

- it is a method to place a **dollar value on intangible benefits.**
- It is also grounded in **welfare economic theory, which embodies patient preferences and choice.**
- **WTP reveals individuals' attitudes towards risk.** How health conscious or motivated are you?
- **Incorporates patients preferences (values)** for use in the decision making process

Disadvantages of WTP

- **Protest response**

- Especially in pre-paid health care system people don't actually pay for health care, **the WTP method can give rise to a lot of protest responses if people misunderstand the task.**

- Some people don't understand the hypothetical nature of the technique and **think that we are using their WTP amount to set charges for health care.**

- They protest by either **saying nothing or zero.**

- **Estimates vary widely**
 - It is **difficult for people to place a dollar value on a health benefit** or an increase in health-related quality of life or satisfaction.
 - Derived on the basis of **what people say rather than do.**

Different techniques of asking WTP

- 1. Open-ended question-** simply ask respondents how much they would be WTP for the program or intervention. Ex. **What is the maximum amount that you would be willing to pay for a 1-hour consultation with a pharmacist?**
- 2. Closed ended question-** Respondents are asked whether or not they will pay a specified dollar amount for the program or intervention. Ex. Would you be willing to pay \$60 for a 1-hour consultation with a pharmacist? ____ Yes ____ No

Different techniques of asking WTP ...

3. **Payment card/scale question-** The payment card method provides the respondent with a list of possible WTP amounts (i.e., payment card) to choose from. Ex. What is the maximum amount that you would be willing to pay for a 1-hour consultation with a pharmacist? Please circle your choice.

\$150 \$90 \$30

\$130 \$70 \$10

\$110 \$50 \$0

Different techniques of asking WTP ...

4. Bidding game question

- several bids are offered **to reach a person's maximum WTP.**
- Before asking a second response, the bids are adjusted based on the first response.
- Ex. Would you be willing to pay \$60 for a 1-hour consultation with a pharmacist?
 - If yes, ask: “Would you be willing to pay \$80?”
 - If no, ask: “Would you be willing to pay \$40?”

CBA...

There are two ways to express the results:

1. Calculate the **Benefit to Cost ratio** for each action- CBA results can be calculated by summing up the total benefits and dividing by the total costs.

Benefit (\$)

Cost (\$)

2. Or calculate the **Net Benefit- calculation simply presents the difference between the total costs and benefits.**

Net benefit= **Benefit (\$)** – **Cost (\$)**

Choose alternative with the largest value.

CBA example

- Cost per flu shot = \$10
- Treatment cost per flu = \$250
- Productivity loss from sick leave = \$4,000
- Employees = 1000
- W/o vaccine: 50 have flu, 3 absence
- W/ vaccine: 30 have flu, 1 absence
- What should the manager do?
 - Should he implement vaccination of all workers?

CBA example

- **Net Benefit** = benefit - cost

$$=(\text{number of flu avoided}) * \$250 + (\text{number of absence avoided}) * \$4000 - \$10 * 1000$$

$$=20 * \$250 + 2 * \$4000 - \$10000 = \$3000 > 0$$

CBA example...

- On the other hand, New flu vaccine available, with
Cost = \$20
- W/ the new vaccine: 5 have flu, no absence from work
- Which one should the manager choose?

Net Benefit= benefit – cost,

$$\text{NB}(\text{new}) = 45 * \$250 + 3 * \$4,000 - \$20 * 1,000 = \$3,250.$$

- $\text{NB}(\text{new}) > \text{NB}(\text{old})$, so that choose the new vaccine

Cost Effectiveness Analysis

Introduction to CEA

CEA:

- **is the most common type of pharmacoeconomic research seen in the literature.**
- **measures costs in dollars and outcomes in natural health units, which indicate an improvement in health such as cures, lives saved, or blood pressure reductions.**

CEA...

- is a method of comparing alternative treatments **in which *the costs and consequences* of the treatments vary.**
- is a method to determine which program or treatment **accomplishes a given objective at the least cost.**
- Measures health benefit by **health outcome, not the dollar value of life**
- the effectiveness is **expressed in terms of non-monetary units that describes the desired objective.**
 - lives saved (life years gained)
 - disability days avoided
 - cases treated

An advantage:

- health units are **common outcomes (clinical units)** that are **routinely measured in clinical trials**, so they are familiar to practitioners. These outcomes do not need to be converted to monetary values.

Limitation:

- CEA cannot be used to compare interventions with different health outcomes because of *its nonmonetary measurement of outcomes*.
- the alternatives used in the **comparison must have outcomes that are measured in the same clinical units.**

For example,

- You cannot use CEA to directly compare **the outcomes of an antihypertensive product** (which may measure mm Hg changes to determine the outcome) **with the outcomes of an asthma product** (which may measure forced expiratory volume [FEV] to determine the outcome).

...Introduction to CEA ...

- For many medications, both the **effectiveness in treating the disease and the side effects of treatment may differ significantly between alternative treatments.**

Example: one chemotherapy regimen may be more effective in lengthening the time until the disease progresses than another chemotherapy regimen, but the more effective regimen may also cause more toxic side effects.

CEA...

- With the CEA method, it is **difficult to collapse different outcomes into one unit of measurement**
- Some researchers consider **cost-utility analysis** (CUA) to be a special subset of CEA that uses units such as **quality-adjusted life years (QALYs)** to collapse different types of outcomes into one unit of measure.

- **Only one outcome at a time can be compared**
- Cost-Effective is **NOT the least expensive!**
- Recommended Criteria, calling an intervention cost effective:
 - Less costly and at least as effective (as other alternatives).
 - More costly and more effective, with **the added benefit worth the added cost**
 - Less effective and less costly, **with the added benefit of the alternative not worth the added cost.**
 - Challenge: Decision as to whether the added benefit worth the added cost

Cost and effectiveness comparison...

- To **determine whether** a therapy or service is **cost-effective**, both the costs and the effectiveness must be considered. Think of comparing a new drug with the current standard treatment.
- The cells represent possible results when comparing two alternatives with regard to costs and effectiveness.

COST-EFFECTIVENESS	Lower Cost	Same Cost	Higher Cost
Lower effectiveness	A Conduct ICER	B Dominated	C Dominated
Same effectiveness	D Dominant	E Arbitrary	F Dominated
Higher effectiveness	G Dominant	H Dominant	I Conduct ICER

Cost and effectiveness comparison ...

- If one compares a new alternative with a standard alternative,
- the lightly shaded cells (B, C, or F) represent when the **new alternative would not be considered** cost-effective (i.e., would be dominated by the standard alternative),
- and the darkly shaded cells (D, G, or H) represent when **the new alternative would be considered** cost-effective (i.e., the dominant choice).

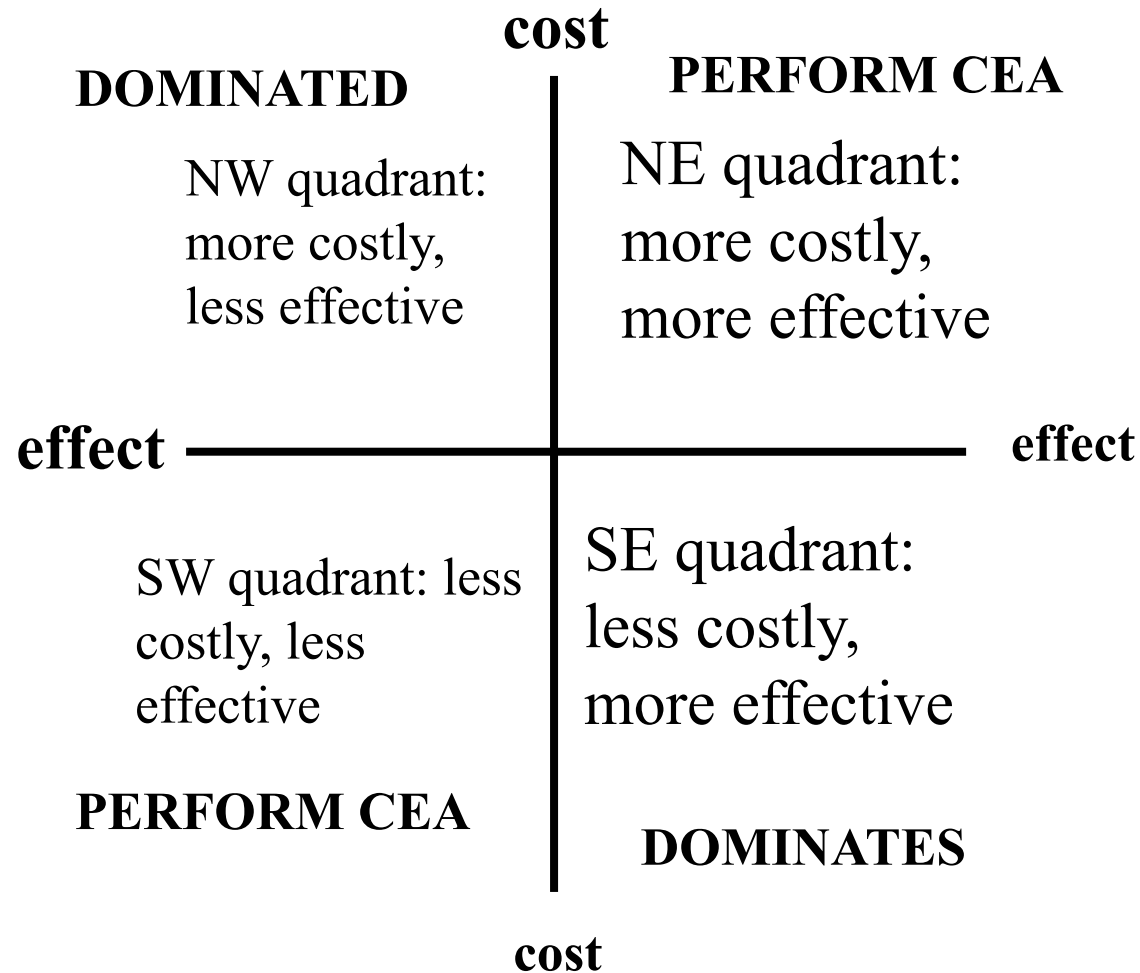
COST-EFFECTIVENESS	Lower Cost	Same Cost	Higher Cost
Lower effectiveness	A Conduct ICER	B Dominated	C Dominated
Same effectiveness	D Dominant	E Arbitrary	F Dominated
Higher effectiveness	G Dominant	H Dominant	I Conduct ICER

- If the comparison falls in the non shaded cells A or I, more information is needed (e.g., how much extra cost per extra unit of outcome, which is determined by conducting an incremental cost-effectiveness ratio).
- If the comparison shows similar effectiveness and similar costs (cell E), other factors may be considered.

COST-EFFECTIVENESS	Lower Cost	Same Cost	Higher Cost
Lower effectiveness	A Conduct ICER	B Dominated	C Dominated
Same effectiveness	D Dominant	E Arbitrary	F Dominated
Higher effectiveness	G Dominant	H Dominant	I Conduct ICER

Cost-Effectiveness Plane

- A graphical depiction of cost- (cost-effectiveness plane) is a visual method for representing the comparison of alternatives.



Cost-Effectiveness acceptability Plane

DIFFERENCE IN COST

New treatment more costly (+)

IV

ΔC

I

The new treatment is less effective and more expensive

The new treatment is more effective and more expensive

New treatment less effective (-)

Differences in effects

New treatment more effective (+)

III

Existing treatment

Worse

better

ΔE

The new treatment is less effective and less expensive

The new treatment is more effective and less expensive

New treatment less costly (-)

CEA...

Steps in conducting CEA

- i. Frame the problem to be analyzed
- ii. Identify the baseline and options to be analyzed
- iii. Identify the outcome measure
- iv. Identify the relevant costs
- v. Construct the decision model
- vi. Analyze and interpret the results
- vii. Perform sensitivity analysis
- viii. Prepare presentation of results

Steps in conducting CEA ...

Step 1. Frame the problem to be analyzed

- Define the study questions
- Determine the perspective of the study
- Time Frame and Analytic Horizon

Step 2. Identify the baseline and options to be analyzed

- **Baseline**
 - Reference point for the analysis
 - Can be placebo, existing treatment or current standard of care
- **Alternative intervention**
 - New drug or intervention

CEA...

Step 3: Identify the outcome measure

- Must be relevant to the study questions
- Can be categorized into:
 - Intermediate outcome
 - Final outcome
- When it is not feasible to collect primary outcomes because of time or monetary resources, **intermediate or surrogate outcomes, such as laboratory measures or disease markers**
- **(e.g., cholesterol levels or blood pressure measurements) are used as proxies or surrogate endpoints.**
- Both positive and negative outcomes should be addressed
 - Positive outcomes: drug's efficacy measure
 - Negative outcomes: ADR and treatment failure

CEA...

Step 4. Identify the relevant costs

- Relevant costs include- cost of intervention, cost of disease prevented, cost of productivity losses averted, and future unrelated health costs.
- Costs to be included depends on the **perspective chosen**.

Step 5. Construct the decision model

- Critical step
- Can use **software, decision tree or done by hand**
- Requires epidemiologic and economic data
- Outcome and economic data
 - RCT
 - Peer reviewed published literatures

CEA...

Step 6. Analyze and interpret the results

- Calculate the summary measure-in CE ratio
- The basic components of a *C/E ratio* are: Numerator: Net cost C and Denominator: Health Outcome
- Cost-effective compared to what?
- The ratio may take either of the following forms:
 - Average C/E ratio (ACER)
 - Ratio of costs to outcomes for a single intervention (strategy A)

Step 6. Analyse and interpret the results...

➤ **Marginal C/E ratio (MCER)**

- Ratio of additional costs to outcomes obtained from one additional unit of an intervention.

➤ **Incremental C/E ratio (ICER)**

- Ratio of additional costs to outcomes obtained when one intervention is compared to the next most effective intervention

6.1 Average Cost Effectiveness Ratio (ACER)

- Ratio of costs to outcomes for a **single intervention** (strategy A)
 - costs in units of money
 - benefits in **natural units**(life years gained...)

$$ACER = \frac{\text{CostStrategy A}}{\text{OutcomeStrategy A}}$$

- Used to evaluate the average cost per health outcome *for a single program*
- Use to allocate resources between *independent* programs
 - For example, Hepatitis B vaccination vs. HIV testing
- Not used to evaluate mutually exclusive (competing) strategies

Average cost effective ratio

- CER is the ratio of resources used per unit of clinical benefit, and it implies that this calculation
- has been made in relation to “**doing nothing**” or **no treatment**.
- In clinical practice, the question is infrequently, “**Should we treat the patient or not?**” or “**What are the costs and outcomes of this intervention versus no intervention?**”

CEA...

6.2 Marginal Cost Effectiveness Ratio

- Marginal cost effectiveness = what's the extra cost to get the extra effectiveness?
- i.e.
$$\frac{\text{Difference in costs}}{\text{Difference in effectiveness}}$$

(within a single program)
- Ratio of *additional* costs to outcomes obtained from **one additional unit** of an intervention.

$$\frac{\text{Cost Strategy A' - Cost Strategy A}}{\text{Outcome Strategy A' - Outcome Strategy A}}$$

- Examines effects of scale (within a single program)
- Does cost per health outcome increase or decrease as *program changes size*?

6.3 Incremental Cost Effectiveness Ratio

- Ratio of additional costs to outcomes obtained when one intervention is compared to the next most effective intervention

$$ICER = \frac{\text{Incremental Cost}}{\text{Incremental Effectiveness}}$$

$$ICER = \frac{\text{CostStrategy B} - \text{CostStrategy A}}{\text{OutcomeStrategy B} - \text{OutcomeStrategy A}}$$

- Incremental Cost= (Cost of program A)-(Cost of program B).
- Incremental Effectiveness
=(Effectiveness of program A) - (Effectiveness of program B).

CEA...

- When comparing 2 therapies, ICER assesses what **the added cost per net effect for alternative therapy would be.**
- ICER is calculated to determine **the extra cost for each extra unit of outcome.**
- If incremental calculations produce **negative numbers**, this indicates that **one treatment, the dominant option, is both more effective and less expensive than the other, dominated option.**

CEA...

- The magnitude of the negative ratio is difficult to interpret, so it is suggested that authors instead indicate which treatment is the dominant one.
- When one of the alternatives is both more expensive and more effective than another, the **ICER is used to determine the magnitude of the added cost for each unit in health improvement.**

CEA...

Example

	Total Cost/ 100 Patients	Lives Saved/ 100 Patients	Average CE Ratio
• Drug A life saved	\$220,000	79	\$2784.81/
• Drug B life saved	\$20,000	78	\$256.41/

Incremental CE ratio

$$\begin{aligned} &= \frac{\text{Cost of drug A} - \text{Cost of drug B}}{\text{Outcomes of drug A} - \text{Outcomes of drug B}} \\ &= \frac{\$220,000 - \$20,000}{79 \text{ Lives} - 78 \text{ Lives}} = \$ 200,000/\text{life saved} \end{aligned}$$

CEA...

More Example.....

Therapy A: costs \$2500 and saves 10 lives, therefore C/E ratio= \$250/life saved

Therapy B: costs \$5000 and saves 15 lives, therefore C/E ratio= \$333/life saved

ICER: $\frac{\$5000 - \$2500}{15 - 10}$ or \$500/life saved

When to Use ICER?

– between mutually exclusive (competing) interventions

CEA...

- Exclusion of Dominated Alternatives
 - If intervention is both *less effective and more costly than the next* most effective alternative- Strongly Dominated and be eliminated.
 - An intervention can also be dominated when its **ICER is higher than that of the next most effective intervention-** Extended or Weak Dominance.

Performing an ICEA

- Rank the alternative treatment options by health benefit (beginning with the one with the lowest benefit).
- Eliminate treatment alternatives that are strictly dominated.
- Calculate the ICER between each treatment option and the next most expensive option.
- Eliminate treatment options that display extended dominance.
- Determine which treatment options have an ICER that is below the cut-off ICER.

CEA...

Incremental Cost-Effectiveness Ratio (ICER)

$$\text{ICER} = \frac{\text{CB} - \text{CA}}{\text{EB} - \text{EA}}$$

- If $\text{CA} > \text{CB}$ and $\text{EA} < \text{EB}$, then B dominates.
- If $\text{CA} < \text{CB}$ and $\text{EA} > \text{EB}$, then A dominates.
- If, however, $\text{CB} > \text{CA}$ and $\text{EB} > \text{EA}$, choice is not obvious. Use CE.

CEA...

Step 7. sensitivity analysis

- Used to test how robust the previous conclusions are when assumptions vary.
 - For example, discount rate, probability of infection, ... etc.
- Conduct sensitivity analysis on the variables that have uncertain values
- How sensitive are the results to different values of key variables.
- “What if” scenarios, e.g.,
 - a. Assume different cost of drug or outcome
 - b. Assume different probability of an outcome

CEA...

Step 8. Prepare presentation of results

Presentation of a CEA should include the following (Haddix et al 2003)

- The study perspective, time frame and analytic horizon
- The study question
- The assumptions used to build the model
- A description of the intervention
- Evidence of the effectiveness of the intervention
- Identification of all relevant costs
 - Inclusion or exclusion of productivity costs
 - Discount rate
 - Results of incremental analysis
 - Results of sensitivity analysis
 - Discussions

CEA...

Example

Program	Averted Cases	Cost
Program A	100	\$ 50.000
Program B	150	\$ 130.000
Program C	110	\$ 170.000

Order by increasing effectiveness

	Averted Cases	Cost
Program A	100	\$ 50.000
Program C	110	\$ 170.000
Program B	150	\$ 130.000

CEA...

- Check for dominance and Eliminate the dominated

	Averted Cases	Increa m. Effec.	Cost	Increm. Cost	Increm. C-E Ratio
Program A	100	-	\$ 50,000	-	
Program C	110	10	\$ 170,000	\$120,000	12,000
Program B	150	40	\$ 130,000 (\$	(\$	-1,000

• *Re-Calculate ICER without dominated programs*

<i>programs</i>	Averted Cases	Increm . Effec.	Cost	Increm. Cost	Increm. C-E Ratio
Program A	100		\$ 50,000		
Program B	150	50	\$ 130,000	\$80,000	

CEA...

- Calculate ratios without dominated programs*

	Averted Cases	Increm. Effec.	Cost	Increm. Cost	Increm. C-E Ratio
Program A	100	-	\$ 50,000	-	-
Program B	150	50	\$ 130,000	\$ 80,000	\$ 1,600

Choose program with Incremental C-E Ratio $\leq$ perceived value (cut –off value)

If perceived value of an averted case and $< \\$ 1,600$, choose Program A;	$\geq$	\$ 500
If perceived value of an averted case $\geq \\$1,600$, choose Program B.	$\geq$	

CEA...

Exercise (Home Take)

Mutually Exclusive Intervention	Total Outcomes (Life years)	Total Costs
Program A	30	\$ 200,000
Program B	10	\$ 50,000
Program C	15	\$ 130,000
Program D	11	\$170,000

Cost Utility Analysis

CUA- introduction

- Cost-utility analysis (CUA) is a method for comparing treatment alternatives that integrates **patient preferences** and **HRQOL**.
- CUA can compare **cost, quality, and the quantity of patient- years**.
- Cost is measured in dollars, and therapeutic **outcome is measured in patient-weighted utilities rather than in physical units**.
- Often the **utility measurement** used is a **quality-adjusted life year (QALY) gained**.

- QALY is a common measure of health status used in CUA, combining morbidity and mortality data.
- Results of CUA are also expressed in a ratio, a **cost-utility ratio (C:U ratio)**.
- Most often this ratio is translated as **the cost per QALY gained** or some other health-state utility measurement.
- The preferred treatment alternative is that **with the lowest cost per QALY** (or other health-status utility).

CUA introduction ...

- CUA is the most appropriate method to use when comparing programs and treatment alternatives:
 - that are life extending with serious side effects (e.g., cancer chemotherapy),
 - those which produce reductions in morbidity rather than mortality (e.g., medical treatment of arthritis), and

- when HRQOL is the most important health outcome being examined.
- CUA is employed less frequently than other economic evaluation methods because of:
 - a lack of agreement on measuring utilities,
 - difficulty comparing QALYs across patients and populations, and
 - difficulty quantifying patient preferences.

- CUA is complex, and **thus can be limited in scope of application from a hospital perspective.**
- Nevertheless, when comparing treatment alternatives where **HRQOL is the most important health outcome being examined, CUA should be considered.**
- CUAs may also be useful **when comparing treatments and outcomes that are very different** (e.g when comparing the treatment of heart disease with prenatal care) because outcomes for both treatments can be summarized into one common unit, such as QALYs

CUA introduction ...

Definition.....

- The QoL component of the QALY is measured using a metric known as a health utility; hence, the term *cost-utility analysis* is used to describe this form of CEA.
- Cost-utility analysis is a special form of cost effectiveness analysis
- CUA tried **to combine the quality and quantity of life in its outcome measures.**
- Sometimes the treatments that extend life the longest are also the most toxic, so **a measure that incorporates both length of life and quality of life** is needed in these cases.

CUA...

- Many health conditions **do not have an impact on patients' length of life, but only on the quality of their life, and CUA may be a good choice** for comparing treatments for these conditions. Examples include conditions such as hearing loss, seasonal allergies, and erectile dysfunction.
- Example: you extend the life of a diabetic pt by 20 years but in reality, these are not full life-years because they have to live with disability.

CUA introduction ...

- **Utility = satisfaction/well-being** - reflects a consumers (weighted) preferences
- **Utility** is based upon some measurement of **quality of life**.
- Consumers (weighted) preferences- **Utility weights are necessarily subjective** - they elicit an individual's preferences for, or value of, one or more health states.
 - Must: 1. Have interval properties (0 to 1)
2. Be 'anchored' at death and 'good health'

CUA...

- A given ‘utility’ is simply a number representing intensity of preference, relative to full health; or full health for age (given a utility of 1) and death (with a utility of 0).
- Utility is a measure of **preference about an outcome** (a health state), giving an indication of the relative value placed on the health state
 - **Scaled 0 (death) to 1 (full health)**

advantage of CUA

- different types of health outcomes and diseases with multiple outcomes of interest can be compared (unlike in CEA) using one common unit such as the QALY.
- incorporates morbidity and mortality into this one common unit without having to determine or estimate the monetary value of these health outcomes (unlike CBA).

disadvantage

- it is difficult to determine an accurate utility or preference weighted value.

Methods used to measure and estimate utility

- To estimate utility weights for various conditions or “health states” between perfect health and death, two broad methods are used to elicit, or generate, these scores: **direct elicitation and indirect elicitation.**
- Direct elicitation methods (**rating scale, standard gamble, and time tradeoff**) are going to be described below.

Steps in Calculating QALYs

To calculate QALYs, the following steps apply:

1. Develop a description of each disease state or condition of interest.
 2. Choose a method for determining utilities.
 3. Choose subjects who will determine utilities.
 4. Sum the product of utility scores by the length of life for each option to obtain QALYs.
- Each of these steps is explained below.

Methods used to measure and estimate utility

Step 1: Develop a Description of each Disease State or Condition of interest

- The description should concisely **depict the usual health effects expected from the disease** state or condition.
- It should include **the amount of pain or discomfort, any restrictions on activities, the time it may take for treatment, possible changes in health perceptions (worry or concern), and any mental changes.**

Step 2: Choose Method for Determining utilities

- The three most common methods for determining preference, or utility, weights are **rating scales (RS)**, **standard gamble (SG)**, and **time tradeoff (TTO)**.
- For each of these methods, a disease state or condition or multiple disease states or conditions are described to subjects who help determine where these disease states or conditions fall between 0.0 (dead) and 1.0 (perfect health).

Methods used to measure and estimate utility

1. Rating scale(RS)

An RS consists of a line on a page **with scaled markings**, somewhat like a thermometer with perfect health at the top (100) and death at bottom (0).

➤ Individuals are asked to indicate where on the line between the best and the worst imaginable health states they would **rate a pre-defined health state**.

➤ Notice that how long a person will be in the disease state may influence the score.

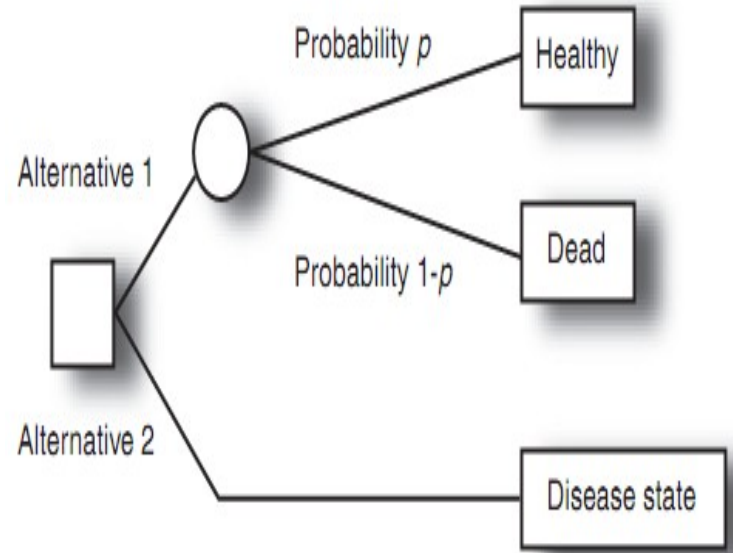
Utility Score	Disease State or Condition
100 (1.0) -	- Perfect health
-	
90 (0.9) -	- Mild allergies
-	- Broken wrist
80 (0.8) -	
-	
70 (0.7) -	- Broken leg
-	
60 (0.6) -	- Hospital dialysis (1 year)
-	
50 (0.5) -	- Hospital dialysis (5 years)
-	
40 (0.4) -	
-	- Severe angina
30 (0.3) -	
-	
20 (0.2) -	
-	
10 (0.1) -	- Coma (1 year)
-	
0 (0.0) -	- Dead

Methods used to measure and estimate utility

2. Standard gamble

- Usually involves asking respondents to identify **what risk they are willing to take to achieve the benefits** of a particular intervention. For this method, each subject is offered two alternatives.

Alternative 1 is treatment with two possible outcomes: **either the return to normal health or immediate death.** **Alternative 2:** is the certain outcome of a chronic disease state for life based on a person's life expectancy



- Classical method of assessing preferences
- Choose between certain outcome and a gamble
- Incorporates uncertainty, therefore better reflects real treatment decisions
- **If respondent is risk neutral then utilities from SG should be the same as from TTO**

Methods used to measure and estimate utility

3. Time trade-off (TTO)

- Again, the subject is offered two alternatives. Alternative 1 is a certain disease state for a specific length of time (t), the life expectancy for a person with the disease, and then death. Alternative 2 is being healthy for time x , which is less than t . Time x is varied until the respondent is indifferent between the two alternatives. The utility score for the health state is calculated as x divided by t .

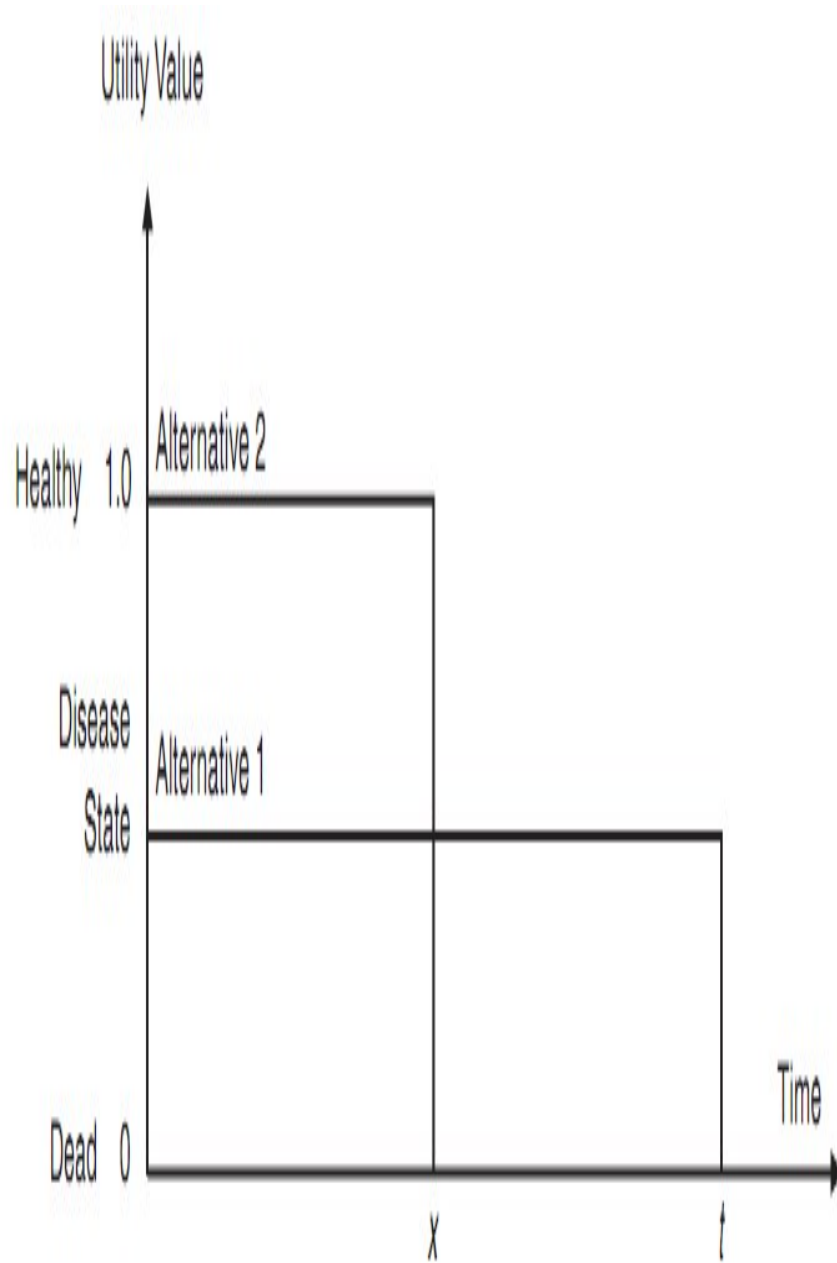
3. Time trade-off (TTO)

- This TTO schematic represents the choice a respondent makes about trading off years of life for better health for a shorter period of time. The respondent is given the choice of living a full life (to time t) with a specific condition or living fewer years (to time x) without the condition (being healthy). The time of living healthy is varied until the respondent is indifferent between living in full health x years and living with the condition for t years. The utility calculated for the condition is x/t .

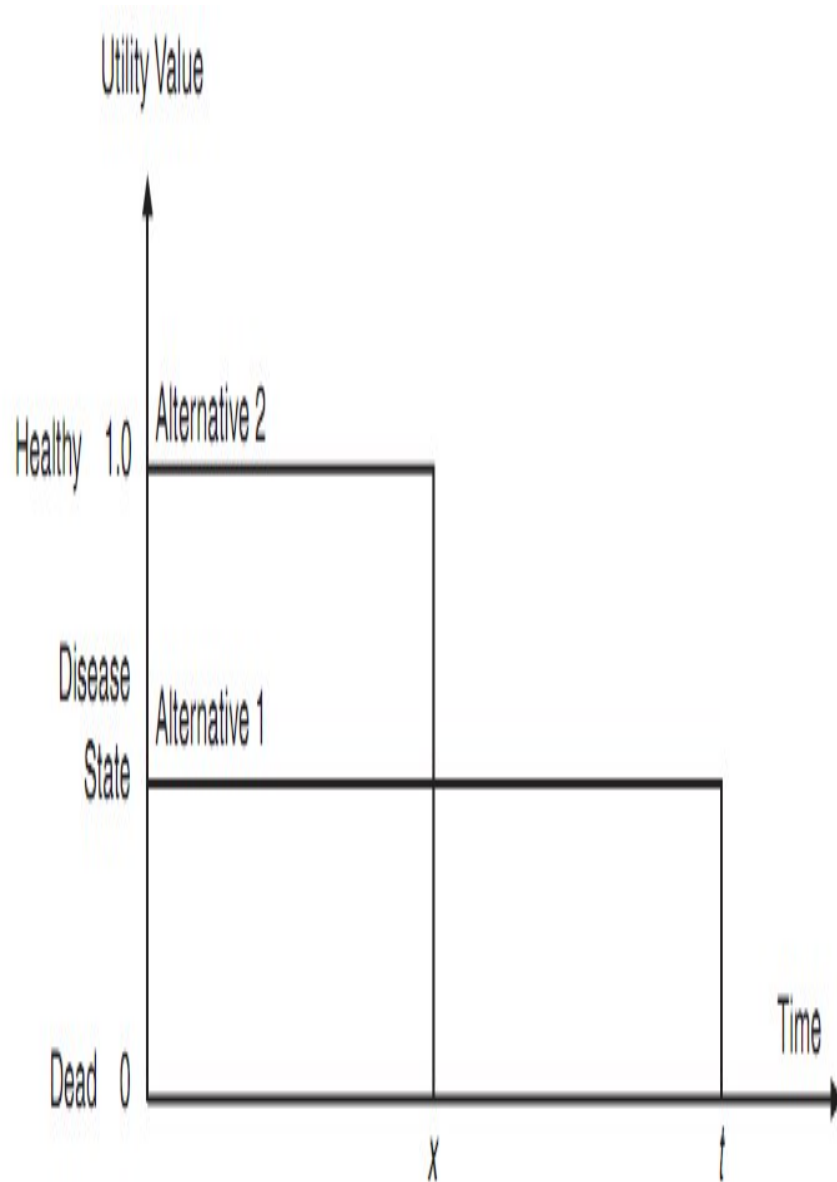
For example, a person with a life expectancy of 40 years is given two options:

Alternative 1 is having a chronic condition (e.g., kidney disease or diabetes) for 40 years, and

Alternative 2 is being healthy (no disease) for 20 years followed by death. If the person says he or she would rather have the disease for 40 years (t) than be healthy for 20 years, the number of years (x) in the healthy state is increased until the person is indifferent between the two alternatives.



If the person would rather be healthy for 20 years than have the disease for 40 years, the number of years (x) in the healthy state is decreased until the person is indifferent between the two alternatives.



Exa: Let us say that for a person who expects to live 40 more years, the person's point of indifference is 30 years of health versus 40 years of kidney disease. The utility score would be $x/t = 30 / 40$ or 0.75.

QALY Calculations

	<i>Cost for Treatment (dollars)</i>	<i>Years of Life Saved</i>	<i>Utility for Each Year of Life Saved</i>	<i>QALYs</i>
Drug A	\$10,000	5	0.8	4.0
Drug B	\$20,000	7	0.5	3.5
	<i>Calculation</i>		<i>Result</i>	
CEA	$(\$20,000 - \$10,000) / (7 \text{ years} - 5 \text{ years})$		\$5,000 per extra year of life	
CUA	$(\$20,000 - \$10,000) / (3.5 \text{ QALYs} - 4.0 \text{ QALYs})$		Drug A dominant	

CUA...

- **When to use CUA**
 - **Health-Related Quality of Life is the important outcome**
 - Programmes affect **both mortality and morbidity** and you want to combine both effects
 - Programmes **affect wide range of outcomes** and you **want common unit for comparison**

When not to use

- Only have **intermediate outcome** data
- **Effectiveness** data show outcomes are **equivalent**
- Effectiveness data show **dominance**
- Extra cost of **obtaining utility values** is **itself not cost-effective** (requires judgment on whether doing so would change result)

Application of CUA

- CUA is the **most difficult and expensive** economic evaluation method to use
- CUA should be used when QoL is the important outcome:
 - **Psychological well-being, physical and social function are important** (eg in the treatment of arthritis)
 - Chemotherapy may increase survival but decrease well-being, **both quality and quantity of life are important**

CUA...

Two limitations of CUA

- Measurement of utility is **very time and resource intensive**.
- **Lack of consensus** on which measurement methods
 - In general, researchers agree that “choice-based” approaches (e.g., standard gamble, time trade-off) are more appropriate.
- **Life Years**
 - Simplest and widely used measure of health
 - Used when the dominant gain from **a change in extra life rather than relief of pain or disability**

Methods of Measurement of outcome in CUA

- Disability adjusted life years (DALY)
- Quality adjusted life years (QALY)

DALY

Disability-Adjusted Life Years (DALYs)

- DALYs for a disease or health condition are **the sum of the years of life lost to due to premature mortality (YLLs) and the years lived with a disability (YLDs)** due to prevalent cases of the disease or health condition in a population.
- Measures **healthy time lost from specific diseases and injuries in a population**
- **Different weights apply to years lived in different ages.**
- The top weight is assigned a year spent at the age of 25 years.

DALY

- DALYs are used by **the WHO to compare population health in different countries.**
- DALYs were developed
 - **To quantify the burden of disease and disability in populations**
 - **To set priorities for resource allocation**

- DALY calculation
 - Calculate **YLL due to each disease** - **Inputs: Life expectancy at age of death, Age at death**
 - Calculate **loss of QoL** of those living with each disease- **Inputs: Duration of disease/injury, Disability weight of disease/injury**
 - Apply weights to reflect social value of people at different ages

Quality of life and Quality-Adjusted Life Years (QALYs)

- “.... Health is physical, mental and social well-being and not merely the absence of disease or infirmity...” – World Health Organization, 1947
- Extending health to well-being: **Quality of life**
- What is the definition of quality of life?
 - Quality of life is **the degree of need and satisfaction within the physical, psychological, social, activity, material and structural area**
- QoL is **subjective evaluation of good and satisfactory character of life as a whole**

QALYs...

- QoL is a state of well-being which is a composite of two components:
 - 1) **The ability to perform everyday activities which reflects physical, psychological, and social well-being**
 - 2) **Patient satisfaction with levels of functioning and the control of disease and/or treatment related symptoms**

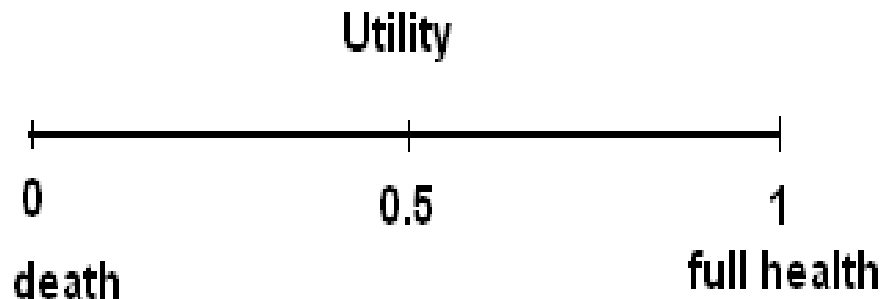
QALYs...

- An individual's perception of their position in life in the context of the **culture** and **values systems** in which they live and in relation to their goals, expectations, standards and concerns (WHO Quality of life Groups, 1993).

QALY

Definition of QALY:- Number of years at full health that would be valued equivalently to the number of life years as experienced.

- The number of QALYs of a person is found by multiplying the expected duration of a health state with its morbidity index and summing up these numbers.
- The utility of an intervention is given by the number of QALYs gained.



- Example: – Persons with permanent kidney failure have lower quality of life, therefore, for these people, 10 years of life might be equivalent to 5 QALYs.
- $\text{QALY} = \text{Utility} \times \# \text{ years in health state}$
- One year at full health $\text{QALY} = 1.0$
- Death $\text{QALY} = 0.0$
- 3 years disabled ($U = 0.5$) = 1.5 QALYs

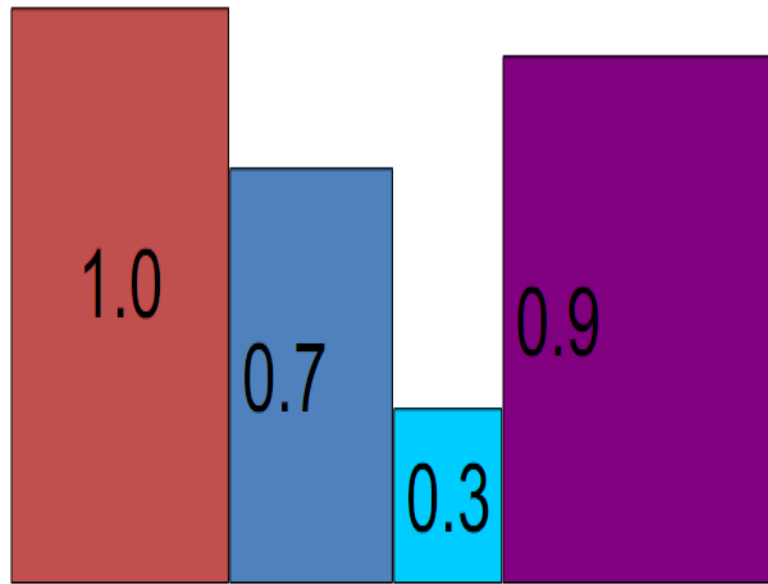
QALY...

QALY Procedure

- Identify possible health states - cover all important and relevant dimensions of QoL.
- Derive ‘weights’ for each state.
- Multiply life years (spent in each state) by ‘weight’ for that state

- Numerical weights
 - Assigned to each possible health state
 - 1 = “perfect health”; 0 = death
 - Reflect strength of preference
- Examples
 - Disability following severe stroke = 0.5
 - Severe angina = 0.7
 - Side effects of beta-blockers = 0.98

QALY...

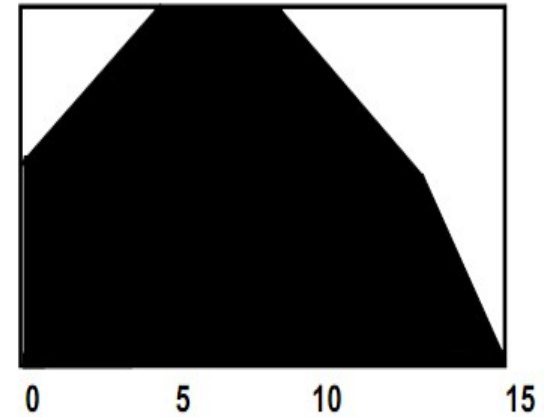


time 0 2 3.5 4.5 7

$$\text{QALYs} = 2 \times 1 + 1.5 \times 0.7 + 1 \times 0.3 + 2.5 \times 0.9 = 5.0$$

Qol Profile

QL Weighting



No Life Years = 15

No QALYs = 11

- Decision Rules:
- CUA result = ICUA, eg cost per QALY gained,
- Decision rule = adopt lowest ICUA
- Question addressed
 1. Should we undertake program “X” or “Y” to treat condition “Z”?
 2. Should we treat condition “A” or “B”?

QALY example

TREATMENT OPTION	TOTAL COST	YEARS OF LIFE	UTILITY	QALY	CU ratio
Drug A	20,000	3.5	0.75	2.6	7692/QALY
Drug B	16,000	2.5	0.8	2.0	8000/QALY

$$\begin{aligned}
 &\text{Example: Incremental CU ratio} \\
 &= \frac{\text{Cost drug A} - \text{Cost drug B}}{\text{QALY drug A} - \text{QALY drug B}} \\
 &= \frac{\$20,000 - \$16,000}{2.6 \text{ QALYs} - 2.0 \text{ QALYs}} \\
 &= \$6,666 / \text{QALY}
 \end{aligned}$$

Example 2

Intervention	Cost (\$)	Effectiveness (Life Expectancy) (y)	Health State (Utility)	QALYs
Treatment A	20,000	4.5	0.60	2.7
Treatment B	10,000	3.5	0.72	2.5

- ICER gained = $20,000 - 10,000 / 4.5 - 3.5 = 10,000$ per LY
- ICUR gained = $20,000 - 10,000 / 2.7 - 2.5 = 50,000$ per QALY

QALY...

Example of QALYs

- a patient with a rare cancer will live for only 2 years without treatment. A new treatment increases life expectancy by 2 years however; it is associated with adverse effects which decrease the quality of life by 25%.

Calculate QALYs

Answer

- Life expectancy = 2 (survival without treatment) + 2 (gain in life years due to treatment) = 4 years.
- Adverse drug reactions due to treatment = 0.25
- Hence decrease in quality of life = $2 \times 0.25 = 0.5$ years.
- Thus net gain is $2 - 0.5 = 1.5$ or 1.5 QALYS.
- Thus the net gain with the new treatment is 1.5 QALYs rather than 2 years.

PE & Formulary Management

- Various issues need to be considered in formulary development.
- Traditionally, formulary decisions took into account **only drug acquisition costs, not the potential savings stemming from use of the better drug**
 - so that **pharmacoeconomics is a preferred tool** with this regard as it takes into account **all types of outcomes.**

Quiz 1

PE-analysis	Cost measurement	Consequence/outcome measurement
CBA		
CEA		
CUA		

THANKS